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Review Article

Deep learning in statistical downscaling for deriving high spatial resolution gridded meteorological data: A systematic review

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ABSTRACT

Nowadays, meteorological data plays a crucial role in various fields such as remote sensing, weather forecasting, climate change, and agriculture. The regional and local studies call for high spatial resolution gridded meteorological data to identify refined details, which however is generally limited due to the models, platforms, sensors, etc. Downscaling has been a significant and practical way to improve spatial resolution. In recent years, with superior feature extraction and expression abilities, deep learning (DL) has outperformed traditional methods in various areas, and exhibits huge potential to establish a complicated mapping between large-scale and local-scale meteorological data. Therefore, this paper provides a systematic review of DL in statistical downscaling for deriving high spatial resolution gridded meteorological data. This review first presents the background, including a taxonomy of downscaling methods, the role of DL in statistical downscaling, and the analogy between downscaling and image super-resolution. It shows evidence of how downscaling can benefit from DL, particularly super-resolution networks. Subsequently, this review focuses on the recent development of the DL-based statistical downscaling of gridded meteorological data, especially the deep architectures, including convolutional neural networks to capture the spatial dependencies of meteorological variables, recurrent neural networks to reveal the temporal states from time series, and generative adversarial networks to facilitate the reconstruction of high-frequency details, as well as the major structure residual learning and attention mechanism. In addition, this review demonstrates the specific issues towards downscaling, including scaling factors, spatial–temporal and variable correlations, and paired datasets construction, and then gives a comprehensive summary of the status of datasets, toolsets and metrics. The future challenges from the perspective of unsupervised models, transformer architecture, data fusion, physical-informed learning, generalization capacity, and uncertainty quantification for downscaling are finally discussed.

1. Introduction

Gridded meteorological data refers to a type of dataset where meteorological variables (e.g., precipitation, surface temperature and wind) are systematically arranged into regularly spaced grids across a geographic region. This data serves a crucial role in a variety of applications such as climate change (Gilabert et al., 2015), quantitative remote sensing (Sun et al., 2020), and agricultural production (Moreira et al., 2019). Primarily derived from in-situ station observations using gauges, satellite remote sensing measurements, and reanalysis products (Yeggina et al., 2020; Zandler et al., 2020; Song et al., 2022), the resulting product is a spatially and temporally complete, gridded dataset of meteorological variables (Sun et al., 2019). Due to the limitations of numerical models, platforms and observation sensors, the spatial resolution of gridded meteorological data typically ranges from tens to hundreds of kilometers (Kikumoto et al., 2015; Maina et al., 2020). The coarse spatial resolution is insufficient for regional or local studies to identify refined details. For example, high spatial-resolution meteorological datasets are required when determining theoretical limitations using radiation budget and energy balance equations (Leng et al., 2017). To fulfill these needs, downscaling, the process to derive high-resolution (HR) fine-scale data from low-resolution (LR) large-scale conditions, is considered a significant way to improve spatial resolutions (Boe et al., 2007; De Caceres et al., 2018).

Downscaling methods for meteorological data mainly include three categories, i.e. dynamic downscaling, statistical downscaling, and the combined methods (Vrac et al., 2012). Dynamical downscaling generally uses simulations of global climate models (GCM) to derive regional

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climate models (RCM) which can generate a set of fine-scale information (Mallard et al., 2018; Tapiador et al., 2020). The resulting simulations are constrained by solid physical principles, but computationally expensive and susceptible to the biases from GCMs (Xu et al., 2019). An alternative is statistical downscaling, which constructs the statistical models that map the relationships between historical large-scale and small-scale meteorological data (Tareghian and Rasmussen, 2013; Dayon et al., 2015). Statistical downscaling offers the advantage of lower computational complexity and ease of application. However, it relies on long-term, high-quality observations to establish a robust statistical relationship (Tang et al., 2016). In addition, there are works to combine the merits of dynamical and statistical downscaling for high accuracy, HR, and physically consistent results with less computational costs (Tran Anh and Taniguchi, 2018; Duchêne et al., 2020; Wang et al., 2020b).

Statistical downscaling has gained wide popularity due to its low computational cost and simplicity. To establish the empirical relationships between coarse and fine-scale variables, various data-driven machine learning (ML) methods, such as multiple linear regression (MLR) (Schoof and Pryor, 2001), support vector machine (SVM) (Chen et al., 2010), random forest (RF) (Davy et al., 2010), and artificial neural network (ANN) (Laddimath and Patil, 2019) have played a significant role in the meteorological service field (Zhang et al., 2022) especially for statistical downscaling. In recent years, with the rapid development of Earth big data and algorithms, a variety of novel methods in Earth monitoring have increasingly emerged (Yuan et al., 2020). Deep learning (DL) has attracted broad attention and outperformed traditional methods in various fields with its superior feature extraction and expression capabilities. The DL models can approximate and capture the complicated non-linear association between meteorological variables, which facilitates understanding for data-driven Earth system science (Reichstein et al., 2019).

Recently, researchers tend to integrate the advantages of DL techniques into spatial downscaling for gridded meteorological data, and they generally follow the ideas of image super-resolution (SR) using DL models. Similar to spatial downscaling that tries to obtain HR small-scale data from LR large-scale data, image SR is generally performed to recover high resolution images from degraded low resolution images resulting from imaging equipment or noise interference, etc. in computer vision (Yang et al., 2019). DL has demonstrated an effective tool for image SR, and a series of DL-based methods have achieved great performance, e.g. super-resolution convolutional neural network (SRCNN) (Dong et al., 2014), enhanced deep residual networks (EDSR) (Lim et al., 2017), meta-transfer learning for zero-shot super-resolution (MZSR) (Soh et al., 2020), and unfolding super-resolution network (USRNet) (Zhang et al., 2020).

However, regardless of the resemblance between downscaling and image SR, applying the existing DL-based image SR networks to the spatial downscaling of meteorological data does not always guarantee good performance. There remain great challenges to be thoroughly considered, e.g. embedding prior knowledge of physical-constrained equations of atmospheric science into downscaling networks, modeling meteorological state spatially and temporally, integrating meteorological variables with complex coupled relationships, and considering the meteorological data distribution. As a result, studies continue to emerge to solve different issues, and various DL-based statistical downscaling methods for gridded meteorological data have been presented.

Currently, there are reviews on dynamical downscaling (Maraun et al., 2010; Wang et al., 2013; Xue et al., 2014; Hong and Kanamitsu, 2014; Giorgi and Gutowski, 2015; Xu et al., 2019; Adachi and Tomita, 2020; Tapiador et al., 2020; Gao et al., 2022) and statistical downscaling (Wilby and Dawson, 2013; Jacobeit et al., 2014; Laddimath and Patil, 2019). These reviews give a detailed introduction to dynamical or traditional statistical downscaling, and the recent development of emerging DL-based methods is not their focus. Most relevant to the topic of this paper, Sdraka et al. (2022) provide an exhaustive review of DL for downscaling RS images. Compared with their intention to obtain high spatial resolution RS imagery by DL-based image fusion and SR, the gridded meteorological data reflecting quantitative features of variables and its corresponding spatial downscaling task in this review possess complexity. Therefore, to the best of our knowledge, this is the first literature review to focus on the statistical downscaling using DL to improve the spatial resolution of gridded meteorological data. It might greatly benefit the refined studies in the fields of remote sensing, meteorology, climatology, etc. Specifically, the main contributions are summarized as follows:

- To give a systematic review of how to use state-of-the-art deep architectures and structures to fulfill downscaling and enhance performance.
- To analyze the similarity and difference between downscaling and image SR, and address the specific issues of scaling factors, spatial–temporal and multi-variable correlations, and paired datasets for downscaling.
- To summarize the status of datasets, available toolsets and evaluation metrics in DL-based downscaling studies. It may be of interest to various data and development needs within the community.
- To provide insights into challenges and future perspectives, including unsupervised model, transformer architecture, data fusion, physical-informed learning, generalization capacity, and uncertainty quantification for spatial downscaling.

The remainder of the paper is organized as follows. Section 2 gives a background introduction to meteorological downscaling, including a method taxonomy, an illustration of DL in statistical downscaling, and the analogy between image SR and downscaling. Section 3 presents the major deep architectures and components in meteorological downscaling networks, and shows how spatial downscaling benefits from DL techniques, especially the DL-based image SR. The endeavors towards the specific issues of spatial downscaling, including downscaling factor, spatial–temporal and multi-variable correlations, and paired datasets are emphatically demonstrated in Section 4. Section 5 summarizes the status of datasets, available toolsets and evaluation metrics, which may be of interest to various demands for the community. Section 6 discusses the future directions, and conclusions are drawn in Section 7.

2. Background

2.1. A taxonomy of downscaling

According to the spatial and temporal dimensions, downscaling can be divided into spatial downscaling and temporal downscaling. Spatial downscaling refers to increasing the spatial resolution of original data, increasing the details of target areas (Jeong et al., 2013), while temporal downscaling is to reduce the time interval between the original data series, e.g. to obtain products with an interval of 6 h or shorter (Paulson and Baxter, 2007). This review focuses on the former. A taxonomy of downscaling methods is shown in Fig. 1, grouped into dynamical downscaling, statistical downscaling, and combined downscaling.

Dynamical downscaling utilizes a HR RCM nested within a GCM to obtain fine-resolution information (Tang et al., 2016). The dynamical downscaling methods developed over the past decades have been reviewed in Xu et al. (2019). They are mainly in four groups, i.e. traditional dynamical downscaling, pseudo global warming, dynamical downscaling with GCM bias corrections, dynamic downscaling with GCM bias corrections and spectral nudging. These methods with physical principles can resolve various processes in climate or Earth systems and interactions, however are computing-intensive and considerably affected by the systematic errors from a GCM which provides the initial and boundary conditions to RCM (Adachi and Tomita, 2020).

For statistical downscaling, the HR fine-scale variables can be downscaled from the larger-scale predictor variables with the established

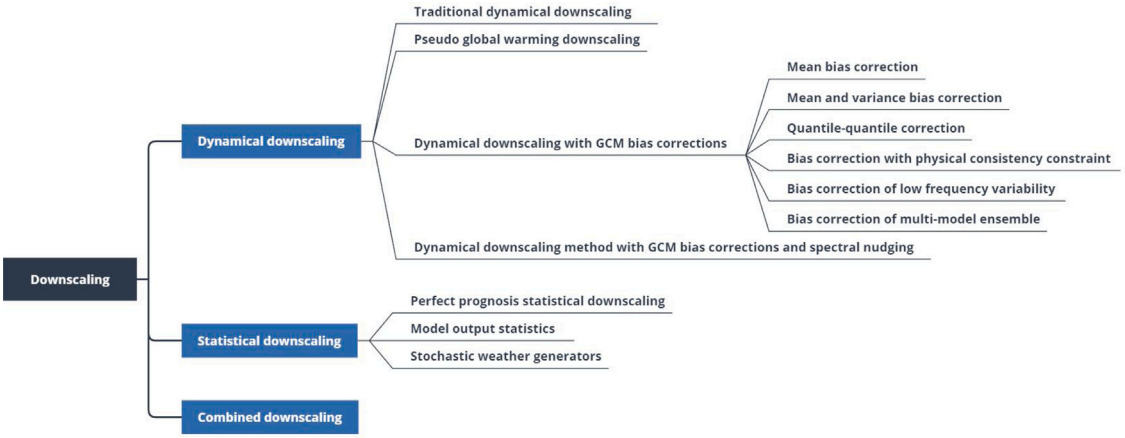


Fig. 1. A taxonomy of downscaling methods divided into dynamical downscaling, statistical downscaling, and the combined downscaling with subcategories (Maraun et al., 2010; Xu et al., 2019). Deep learning works to establish the statistical relationships between variables at large scales and local scales.

statistical mapping from available past data (Jacobeit et al., 2014). It can significantly reduce the computation complexity and is easily configured (Xu et al., 2019). Wilby and Wigley (1997) classified statistical downscaling into regression methods, weather type approaches, and stochastic weather generators (SWGs). Later, Maraun et al. (2010) summarized the statistical downscaling approaches into perfect prognosis (PP), model output statistics (MOS), and SWGs based on the nature of the chosen predictors, as shown in Fig. 1. PP includes the classical regression models and weather pattern-based approaches, if the predictors are realistically simulated. Meanwhile, MOS is to establish a statistical mapping between variables simulated by RCM and local-scale observations to correct RCM errors, and SWGs generate local-scale time series resembling the statistical properties of observed weather.

There are also attempts to combine the advantages of statistics and dynamics in recent years. For example, Tran Anh and Taniguchi (2018) developed a dynamic-statistical model, which downscaled the output of weather research and forecasting (WRF) with ANN further. Han et al. (2019a) proposed a multivariate HR model that combined dynamical downscaling and statistical downscaling. Duchêne et al. (2020) proposed a method combining statistics and dynamics to downscale the results of RCM to the city level. These methods try to provide computation-efficient and accurate downscaling results.

2.2. Deep learning for statistical downscaling

Deep learning and its broader domain machine learning are mainly used in statistical downscaling to establish linear or nonlinear models to relate large-scale predictors to local-scale variables. The downscaling scheme consists of two key steps. The first step selects and transforms the well-informed predictors into a useful feature space, and the second step establishes and develops a statistical model to link large-scale predictors and local-scale predictand for downscaling (Maraun et al., 2010).

Linear models have been widely used for statistical downscaling. Typical linear methods involved include MLR (Zeng et al., 2011; Jia et al., 2011; Fu et al., 2020; Martin et al., 2021), canonical correlation analysis (CCA) (Skourkeas et al., 2013; Wu et al., 2016; Chen et al., 2017), empirical orthogonal function (EOF) (Davy et al., 2010; Martinez et al., 2013) etc. To capture the complex nonlinear relationships between predictors and predictands, nonlinear models have been developed, e.g. random forest (Yan et al., 2021; Goutham et al., 2021; Sulaiman et al., 2022), SVM (Chen et al., 2012; Srinivas et al., 2014; Sachindra et al., 2018; Sa'Adi et al., 2020; Jin et al., 2021; Sulaiman et al., 2022), ANN (Hassan et al., 2015; Sachindra et al., 2018; Nourani et al., 2019; Laddimath and Patil, 2019; Trinh et al., 2021; Martin et al., 2021; Sulaiman et al., 2022).

DL, as a subdomain of ML, has been considered a promising tool for establishing statistical downscaling models recently, due to its superior capability in learning complex relationships between coarse-resolution predictors and fine-resolution target variables. The corresponding problem formulation can be stated as follows. Given a set of LR predictor variables x which can be with auxiliary information, DL networks aim to learn a nonlinear mapping $f^{\omega}(x)$, with a set of parameters ω, that links to the fine-resolution predictand variable y as shown in Eq. (1).

$$y = f^{\omega}(x, \delta), \tag{1}$$

where δ represents the configuration for the downscaling model, e.g. the magnifying scale and region, controlling the output result. The set of model parameters ω is updated to fit the data according to a loss function.

2.3. Statistical downscaling and super-resolution

An example of LR and the corresponding downscaled HR meteorological variable, wind speed are presented in Fig. 2. Its purpose is quite similar to the image SR task in computer vision. It is often the fact that the spatial resolution of images is limited because of the imaging equipment or noise interference. Image SR refers to reconstructing the HR image that contains improved visual perception and enhanced details from the blurred LR image (Lepcha et al., 2023). Given a LR image I_{LR}, it is assumed to be degraded from a HR image I_{HR}. The degradation model Ψ between the HR and LR images can be stated as Eq. (2), while image SR is the inverse process of the degradation model and can be written as Eq. (3). σ represents the parameters of downsampling operations, and θ represents the parameters of SR model F (Wang et al., 2022a).

$$I_{LR} = \Psi(I_{HR}, \sigma). \tag{2}$$

$$I_{SR} = \Psi^{-1}(I_{LR}, \theta) = F(I_{LR}, \theta). \tag{3}$$

The image SR and DL-based statistical downscaling denoted by Eq. (3) and (1) are in a similar form, and are ill-posed inverse problem (Lepcha et al., 2023). Therefore, existing DL-based statistical downscaling methods almost follow the ideas of SR networks, and deal with the challenges considering the characteristics of meteorological data and downscaling tasks. The subsequent sections discuss the DL-based downscaling methods in detail.

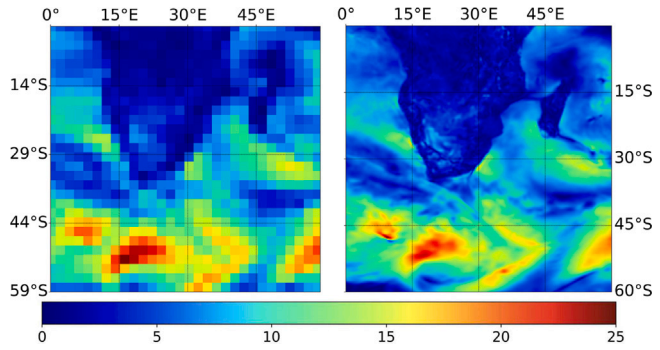


Fig. 2. An illustration of the LR meteorological variable (left) and the HR spatial downscaled result (right), taking wind speed as an example. The colors indicate the values of wind speed (m/s) in the legend. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Deep architectures in downscaling

Compared with traditional neural networks, DL has a deeper number of layers with more complex structures, and can automatically extract and express features hierarchically. According to whether paired images of LR and HR are required during training, the DL methods are classified as supervised and unsupervised methods. From the perspective of network architecture, representative DL networks developed at present include auto-encoder (AE) (Tolstikhin et al., 2017), deep belief network (DBN) (Hinton et al., 2006), deep Boltzmann machine (DBM) (Salakhutdinov and Hinton, 2012), convolutional neural network (CNN) (Aghdam and Heravi, 2017), recurrent neural network (RNN) (Gal and Ghahramani, 2016), generative adversarial network (GAN) (Goodfellow et al., 2014), etc. Currently, the basic architectures of DL-based meteorological data spatial downscaling mainly cover CNN, RNN and GAN. A summary of the DL-based downscaling methods with the corresponding fundamental networks, as well as the status of variables and code are listed in Table 1. The following subsections depict the architectures of recent researches.

3.1. Convolutional neural network

CNN, originally inspired by biological visual perception mechanism, is one of the most widely used architectures at present. It is composed of input layer, convolutional layer, pooling layer and fully connected layer, with the characteristics of parameter sharing and local connection. CNN applies the convolutional layers to extract features of the spatial structure information of inputs, and has fewer parameters compared with the fully connected neural network in the same condition (Zhang et al., 2021c). The superior feature extraction ability of CNN has solved a variety of conventional image processing problems with improved performance and efficiency (Mishra et al., 2022). As reported in Table 1, CNN is the most general architecture of DL-based meteorological data downscaling methods. The data are fed into CNN, which typically contains upsampling layers, convolutional layers, residual modules, etc. for feature extraction and mapping, constituting a typical end-to-end learning.

A milestone is DeepSD (Vandal et al., 2017), the earliest DL-based downscaling network. It was built on the early image SR network SRCNN (Dong et al., 2016a). SRCNN proposed to end-to-end learn the mapping from LR to HR images. It includes three parts, feature extraction module, non-linear mapping module, and reconstruction module, which all consist of convolutional layers. The first module extracts the features of LR images, and obtains a series of feature maps; the non-linear mapping module constructs the mapping between feature maps and HR images; the reconstruction module processes the HR features output by the mapping module through a simple convolutional layer to obtain the final SR outputs. It completes the SR task in one network, and performs better than the previous step-by-step model. DeepSD augments SRCNN with multiple variable input channels, i.e. the precipitation with auxiliary topographical elevation to enhance predictability in statistical downscaling. Specifically, an architecture of stacked SRCNNs has been designed for DeepSD, which allows the model to learn multi-scale patterns, requiring less complexity in the spatial representations (Vandal et al., 2017). Similarly, Kumar et al. (2021) employed three DL-based algorithms derived from SRCNN, i.e. SRCNN, stacked SRCNN, and DeepSD for downscaling of summer monsoon rainfall data over India. Later, Passarella et al. (2022) selected the fast super-resolution convolutional neural network (FSRCNN) (Dong et al., 2016b) as the base model, improved it by using additional SRCNN-like convolutional layers after the deconvolution step to extract finer spatial details, and obtained the resulting FSRCNN-ESM. Fig. 3 shows a representative evolution from the image SR network to the corresponding downscaling network, from SRCNN to DeepSD (a) and from FSRCNN to FSRCNN-ESM (b), as examples.

After DeepSD, a series of CNN-based spatial downscaling methods were proposed. Some work tries to design customized downscaling networks. For example, Pan et al. (2019) introduced CNN for daily precipitation prediction, and considered the physical principles by restricting the predictors to the variables directly resolved by discretizing the atmospheric dynamics equations. Mu et al. (2020b) proposed a climate downscaling network (CDN) by considering the multi-scale spatial correlations and chaos of climate events, and incorporating the priori meteorological knowledge. Wang et al. (2021) presented a super-resolution deep residual network (SRDRN) with the incorporation of residual blocks, batch normalization, and data augmentation for downscaling daily precipitation and temperature. Rampal et al. (2022) considered the complexity of network architecture, loss functions, input domain size, and training data amount, and designed simple CNNs for downscaling historical rainfall in the highly complex terrain setting of New Zealand. Dujardin and Lehning (2022) presented Wind-Topo, an approach with multiple CNN branches to reveal the interactions between HR topography and LR states of atmosphere to generate HR near-surface wind fields. Jin et al. (2022) proposed a simple CNN model with general circulation fields as inputs, and further applied a transfer learning strategy for seasonal precipitation prediction over China. In this category of studies, a set of representative CNN-based models such as CNN1, CNN10, CNN-PR, and CNNdense have been developed and made freely available from Bano-Medina et al. (2020). These methods were later thoroughly evaluated in Bano-Medina (2020), Bano-Medina et al. (2021), and Sun and Lan (2021).

In addition, a majority of methods evolve from the existing semantic segmentation networks (e.g. UNet). UNet is a symmetrical encoder–decoder CNN for semantic segmentation, which consists of a contracting path to capture context and a symmetric expanding path that enables precise localization (Ronneberger et al., 2015). A variety of researches modified UNet to learn complex semantic features from inputs, reconstruct HR by the hierarchical decoder, and be better adapted to downscaling as a regression problem. Representatives include deep residual U-Net (DeepRU) for wind downscaling (Höhlein et al., 2020), UNet-autoencoder (UNet-AE) (Sha et al., 2020a), Nest-UNet (Sha et al., 2020b) and RCM-emulator (Doury et al., 2022) for temperature downscaling, an attention-based U-Net model (AU-Net) (Sun and Tang, 2020) and a modified climate4R (Quesada-Chacon et al., 2022) for precipitation downscaling. Fig. 4 shows how UNet is modified into a downscaling network, taking UNet-AE as an example (Sha et al., 2020a).

Similar to DeepSD, apart from semantic segmentation networks like UNet, recent studies largely benefit from state-of-the-art image SR Networks, e.g. the laplacian pyramid super-resolution network (LapSRN) with residual dense block (ResLap) (Cheng et al., 2020), China Meteorological Administration's land assimilation system statistical downscaling model (CLDASS) (Tie et al., 2022) based on very deep super-resolution network (VDSR) (Kim et al., 2016), FSRCNN-ESM (Passarella et al., 2022) based on FSRCNN (Dong et al., 2016b), a DL

Table 1
A summary of deep learning-based meteorological data downscaling methods, including the network structure types, fundamental networks, meteorological variables studied, the max scale that refers to the maximum factors increasing the resolution, and the code accessibility.

Type	Spatial downscaling network	Fundamental network	Meteorological Variables	Max scale	Code
CNN	DeepSD (Vandal et al., 2017)	SRCNN (Dong et al., 2016a)	Precipitation	8	✓
	Customized CNN (Pan et al., 2019)	/	Precipitation	25	×
	CDN (Mu et al., 2020b)	/	Precipitation	5	×
	SRDRN (Wang et al., 2021)	/	Temperature, precipitation	24	×
	Wind-Topo (Dujardin and Lehning, 2022)	/	Wind	22	×
	Non-Linear CNN (Gamma), Linear CNN (Rampal et al., 2022)	/	Rainfall	10	×
	Customized CNN (Jin et al., 2022)	/	Precipitation	4	×
	convCNPs (Vaughan et al., 2022)	/	Temperature, precipitation	–	✓
	CNN1, CNN10 (Bano-Medina, 2020)	/	Temperature, precipitation	4	✓
	CNN1, CNN10, CNN-PR, CNNdense (Bano-Medina et al., 2020)	/	Temperature, precipitation	4	✓
	CNN1, CNN10 (Bano-Medina et al., 2021)	/	Temperature, precipitation	4	✓
	CNN1, CNN10, CNN-PR, CNNdense (Sun and Lan, 2021)	CNNs (Bano-Medina et al., 2020)	Temperature, precipitation	6.67	✓
	DeepRU (Höhlein et al., 2020)	UNet (Ronneberger et al., 2015)	Wind	3.44	×
	UNet-AE (Sha et al., 2020a)	UNet (Ronneberger et al., 2015)	Temperature	6.25	×
	Nest-UNet (Sha et al., 2020b)	UNet (Ronneberger et al., 2015)	Precipitation	6.25	×
	RCM-emulator (Doury et al., 2022)	UNet (Ronneberger et al., 2015)	Temperature	12.5	✓
	AU-Net (Sun and Tang, 2020)	UNet (Ronneberger et al., 2015)	Precipitation	25	×
	Modified climate4R (Quesada-Chacon et al., 2022)	UNet (Ronneberger et al., 2015), UNet++ (Zhou et al., 2020)	Precipitation	25	✓
	ResLap (Cheng et al., 2020)	LapSRN (Lai et al., 2017)	Precipitation	5	×
	Stacked SRCNNs, DeepSD (Kumar et al., 2021)	SRCNN (Dong et al., 2016a)	Rainfall	4	×
	SPCN (Jiang et al., 2021)	ESPCN (Shi et al., 2016)	Precipitation	8	×
	CLDASSD (Tie et al., 2022)	VDSR (Kim et al., 2016)	Temperature	6.25	×
	FSRCNN-ESM (Passarella et al., 2022)	FSRCNN (Dong et al., 2016b)	Temperature, heat flux, precipitation	4	×
	DeepCAMS (Xiao et al., 2022)	ABPN (Liu et al., 2019b)	$PM_{2.5}$	3	✓

(*continued on next page*)

Table 1 (*continued*).

Type	Spatial downscaling network	Fundamental network	Meteorological Variables	Max scale	Code
RNN	BiGRU (Zhang and Li, 2021)	GRU (Chung et al., 2014)	Wind	6	×
	TRU-NET (Adewoyin et al., 2021)	ConvGRU (Shi et al., 2017)	Precipitation	7	✓
	Convolutional LSTM (Chou et al., 2021)	LSTM (Graves et al., 2013)	Precipitation	5	×
	Augmented-ConvLSTM (Harilal et al., 2021)	LSTM (Graves et al., 2013)	precipitation	4.6	✓
GAN	PhIRE (Stengel et al., 2020)	SRGAN (Ledig et al., 2017)	Wind, solar	50	✓
	DeepDT (Cheng et al., 2022)	ESRGAN (Wang et al., 2019a)	Precipitation	10	×
	SGAN (Leinonen et al., 2021)	/	Precipitation, cloud optical thickness	16	✓
	Conditional GAN, VAE-GAN (Harris et al., 2022)	Recurrent, stochastic SR GAN (Leinonen et al., 2021)	Precipitation	10	✓
	GDRN (Liu et al., 2022)	DRN (Guo et al., 2020)	Sea surface wind	16	×

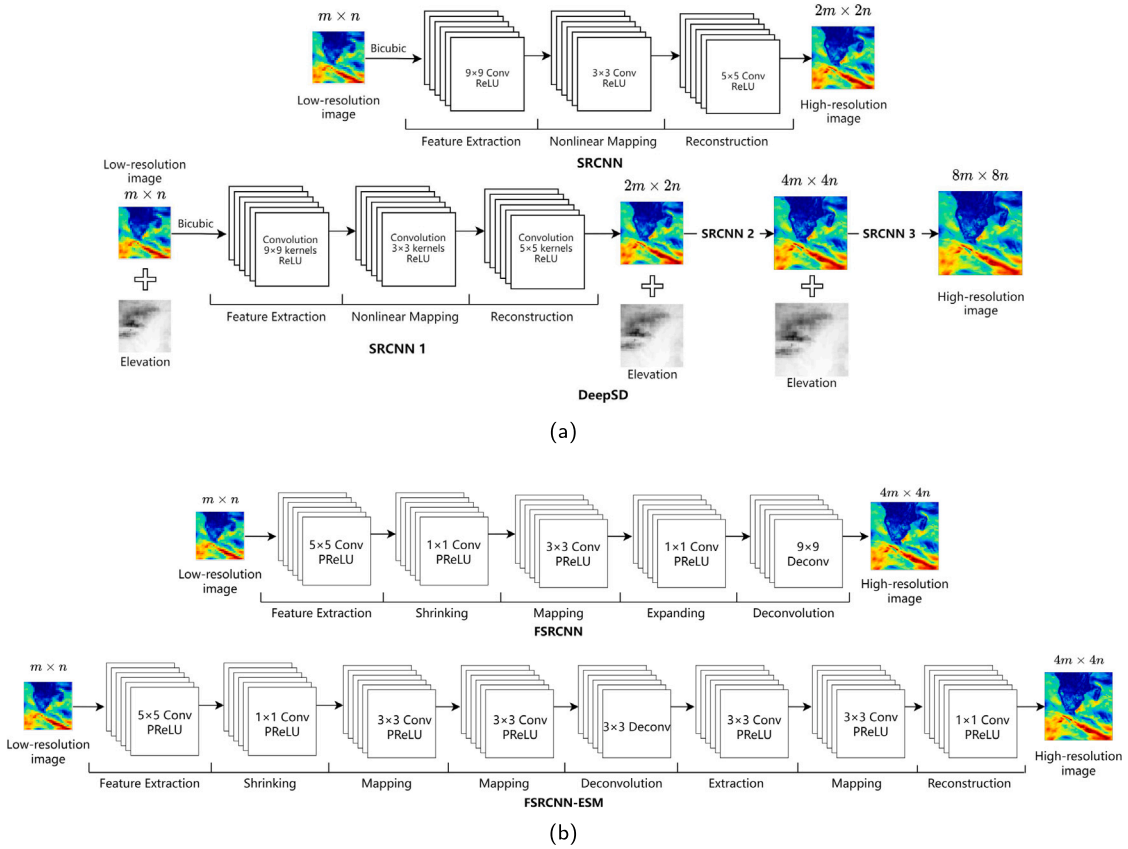


Fig. 3. Representative examples of how existing the CNN-based image SR networks evolve into downscaling networks: (a) from SRCNN (top) (Dong et al., 2016a) to DeepSD (bottom) (Vandal et al., 2017), and (b) from FSRCNN (top) (Dong et al., 2016b) to FSRCNN-ESM (bottom) (Passarella et al., 2022). These networks are generally composed of feature extraction, nonlinear mapping, reconstruction, and deconvolution modules.

framework to downscale Copernicus Atmosphere Monitoring Service (DeepCAMS) (Xiao et al., 2022) with attention based back projection network (ABPN) (Liu et al., 2019b).

Most of the existing downscaling networks directly link predictors and predictands, and the trained models are used to yield the HR outputs at a discrete set of locations. Instead, a few studies integrated CNN as a major component. For instance, Vaughan et al. (2022) developed a convolutional conditional neural process (convCNP) that outputs a stochastic process to support queries at an arbitrary latitude–longitude coordinate. For convCNP, CNN is taken as the first step to extract features and predict variables, followed by the translation to off-the-grid predictions, and inclusion of sub-grid-scale topography. Jiang et al.

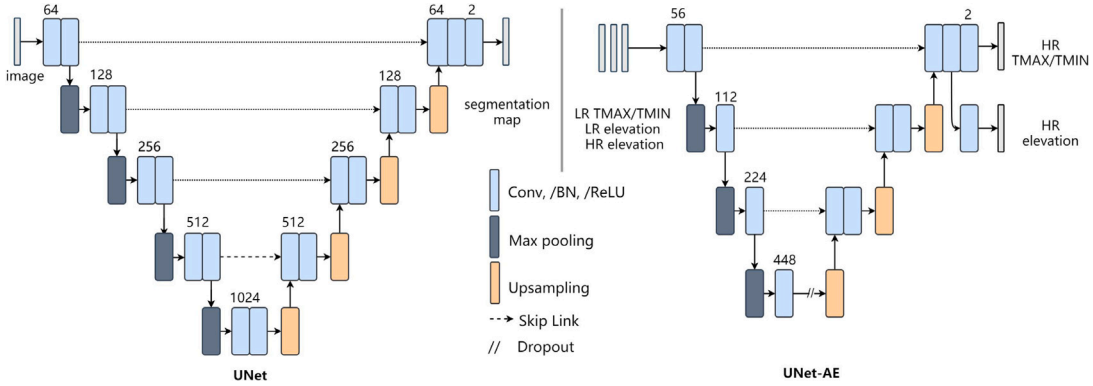


Fig. 4. Representative examples of how existing semantic segmentation networks evolve into downscaling networks: from UNet (left) (Ronneberger et al., 2015) to UNet-AE (right) (Sha et al., 2020a). In general, the downward module is called the encoder, and the upward module is called the decoder. Among them, the blue modules represent the convolution layer with optional batch normalization and ReLU blocks, and TMAX/TMIN refers to the maximum or minimum temperatures respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(2021) developed an approach to construct long-term HR precipitation over complex-terrain areas by correcting the ERA5 precipitation using the HR simulation, and then performing downscaling with a CNN model based on efficient sub-pixel convolutional neural network (ESPCN).

3.2. Recurrent neural network

The CNN-based models attempt to capture the complicated spatial dependencies of variables and anomalies, while the variability also exhibits memory in time, i.e. the present state of the system may influence the future states (Harilal et al., 2021). Therefore, RNN has been integrated to model the spatio-temporal dependencies simultaneously, thus enhancing the downscaling performance. RNN is an architecture for the sequential or time-series fields (Gal and Ghahramani, 2016). Assuming that the input and output are not independent of each other, RNN contains non-linear recurrent structures. It makes the state of hidden layers affected by the input data and the previous calculation results (Qiu et al., 2023). RNN has been considered a significant tool for non-linear time-series forecasting, and typical ones include gated recurrent unit (GRU) (Chung et al., 2014; Luo et al., 2021), long short-term memory networks (LSTM) (Graves et al., 2013) etc. The previous studies for the meteorological data downscaling with RNNs are summarized in Table 1. Fig. 5 shows the schematic diagrams of using bidirectional gated recurrent unit (BiGRU) (Zhang and Li, 2021) and augmented LSTM (Harilal et al., 2021) for downscaling.

Specifically, considering the temporal dependence and the interaction between meteorological factors, Harilal et al. (2021) proposed ConvLSTM-SR, a downscaling model based on augmented LSTM. The network consists of three ConvLSTM modules and a SR module, shown in Fig. 5(a). ConvLSTM adopts the LSTM structure made up of convolution operations, so that it can retain the spatial characteristics of input data. The SR module consists of four stacked convolution layers, with a skip connection. It is followed by two lightweight convolution layers, through which the output of SR is reconstructed to obtain the HR downscaling result. Similarly, Chou et al. (2021) also developed a SR-based ConvLSTM downscaling network to generate high spatial resolution monthly precipitation data in China. Adewoyin et al. (2021) integrated these recurrent structures in an encoder–decoder model, and presented a temporal recurrent U-Net (TRU-NET) to produce HR precipitation. They adopted the UNet architecture, and incorporated the ConvGRU layers (Shi et al., 2017) into the encoder and decoder of UNet with a novel fused temporal cross attention mechanism, thus utilizing the temporal information present in weather data and modeling multiple temporal scales of weather dynamics.

Furthermore, taking into account that both the past and future features may have impacts on the current status, Zhang and Li (2021) proposed a spatial downscaling method for sea surface wind speed with BiGRU, which can efficiently use the past and future features by the bidirectional hidden layers and the gate units (Jabreel and Moreno, 2017), as shown in Fig. 5(b). The trained BiGRU model with ERA-Interim (Dee et al., 2011) and ERA5 (Hoffmann et al., 2019) reanalysis products has been applied to downscale the results of Coupled Model Intercomparison Project Phase 6 (CMIP6) wind speed simulations.

3.3. Generative adversarial network

GAN is a self-supervised learning model that is composed of two parts, generator and discriminator. The generator is used to generate data, and the discriminator is to discriminate whether the data comes from the generator or real samples. GAN has been introduced into the image SR (Goodfellow et al., 2014), and further applied to meteorological data downscaling, to solve the problem that the current results tend to be smooth with fewer high-frequency details. Fig. 6 gives a schematic using the GAN architecture for meteorological downscaling.

A representative work is the adversarial super-resolution method (PhIRE) (Stengel et al., 2020), which can downscale wind field by 50x based on the super-resolution generative adversarial network (SRGAN) (Ledig et al., 2017). The network is made up of a generator and a discriminator, and through supervised learning of the generator and the adversarial learning of both, the results with higher frequency details can be obtained. The generator applies the two stacked SR networks (10x and 5x, respectively) in SRGAN to generate high factor downscaling results. PhIRE can achieve better results by introducing the additional perceptual loss from adversarial learning. Likewise, Cheng et al. (2022) also remodeled from SRGAN (Ledig et al., 2017) and proposed DeepDT, a GAN-based method for downscaling precipitation. They generated the input data with a 1×1 Conv layer that fused different variables, i.e. precipitation, topography, temperature, and humidity. Compared to SRGAN, the fusion of multiple variables and simplified perceptual loss make DeepDT construct high-quality data for precipitation prediction.

Additionally, there are studies that demonstrate significant efforts to combine various DL network architectures for downscaling. For instance, Leinonen et al. (2021) introduced a recurrent, stochastic SR GAN that can generate ensembles of time-evolving HR atmospheric fields from LR image sequence inputs. The proposed network embedded recurrent structures in the form of ConvGRUs in both the generator and discriminator, which permits the network to learn the temporal evolution of data. Moreover, the generator contains a noise input using a different noise instance for each time step, updated by the recurrent layers, so that the generator can learn to map the variability of the output or represent uncertainty in meteorological data problems. Later, Harris et al. (2022) extended the approach in Leinonen et al.

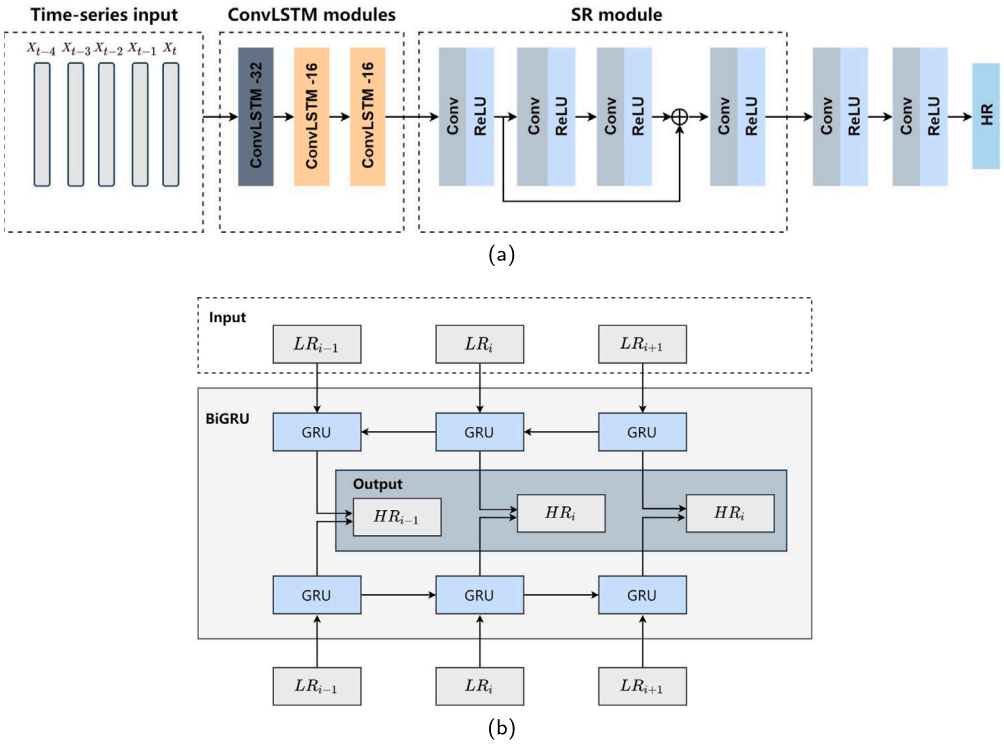


Fig. 5. Representative examples of the schematic diagrams using RNNs for downscaling: (a) an augmented LSTM downscaling network (Harilal et al., 2021), and (b) a BiGRU based downscaling network (Zhang and Li, 2021). The inputs of these downscaling networks are time series data, and the input X_t is composed of multiple meteorological variables.

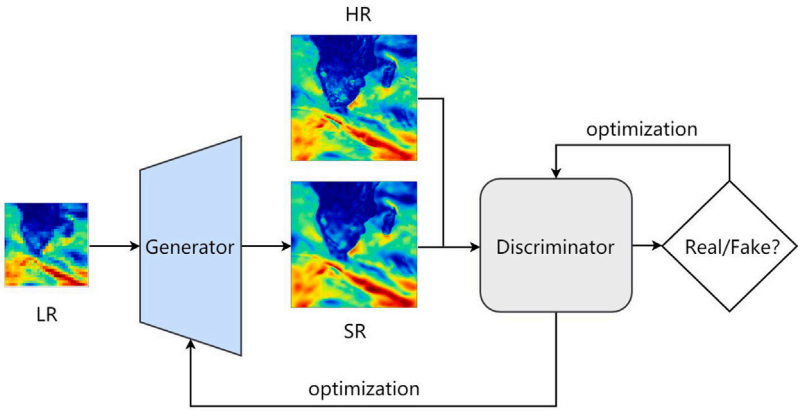


Fig. 6. A schematic of using GAN for downscaling, where discriminator is trained to discriminate if the input is from HR or downscaled SR, while generator is to generate SR from LR and can be additionally optimized with the supervision of discriminator.

(2021) by directly constructing the mapping from multiple LR atmospheric fields originating from a weather forecast model to HR "truth" radar data, and can realize the resolution increase and error correction. They used two generative DL models, a GAN-based model, and a conditional variational auto-encoder GAN (VAE-GAN) based model. Multiple physical principles-related variables, including topography data and a land–sea mask, are also incorporated.

Considering the challenge of the absence of HR ground truths, Liu et al. (2022) proposed an unsupervised network based on generative adversarial networks and dual learning, marked as GDRN. The proposed method performs a primal task to reconstruct HR images and a dual task to estimate the degradation kernels in the generator, and provides a solution for the downscaling without unpaired data by a model adaptation strategy. The experiments show better downscaling accuracy with more fine-grained details in terms of reconstruction quality.

3.4. Other major structures

This subsection presents other critical structures generally used in the meteorological downscaling networks to enhance performance, including residual modules and attention mechanisms.

3.4.1. Residual learning

With the increase in network depth, degradation is a significant issue for model training. He et al. (2016) proposed residual learning to ease the training of deep networks. The residual network adds a skip connection to a normal layer. Adding the input of a layer to the output is equivalent to offering a shortcut for the layer.

Residual learning has been widely adopted in image SR networks, e.g. residual dense network for super-resolution (RDN) (Zhang et al., 2018b), very deep convolutional networks (VDSR) (Kim et al., 2016), enhanced deep residual networks for super-resolution (EDSR) (Lim

et al., 2017). It mainly includes two strategies, global residual learning (GRL) and local residual learning (LRL) (Sdraka et al., 2022). GRL is employed to learn the residual between the input and output images to recover the high-frequency details, while LRL represents a local shortcut inserted between intermediate layers to alleviate vanishing gradients along with the increasing network depth (Lepcha et al., 2023).

Meteorological downscaling networks also take advantage of global and local residual structures to facilitate training and improve performance. Wang et al. (2021) proposed SRDRN for the spatial downscaling of daily maximum and minimum temperature and precipitation, which contains the same residual blocks (RB) in SRGAN (Ledig et al., 2017). Cheng et al. (2020) developed an augmented method ResLap based on LapSRN (Lai et al., 2017), which provided continuous memory and local feature fusion mechanisms, and also introduced a novel residual dense block, improving the connection between features in different layers. More works with LRL can be found in Harris et al. (2022), Liu et al. (2022), Stengel et al. (2020), Cheng et al. (2022), and Leinonen et al. (2021). In addition, Höhlein et al. (2020) proposed DeepRU, a U-shaped deep residual network based on the UNet architecture, which can achieve great downscaling performance in deeper networks through the residual structure and skip connections at multiple resolution stages. Similarly, the works in Wang et al. (2021) and Tie et al. (2022) also applied both global skip connection and local residual blocks.

3.4.2. Attention mechanism

Motivated by that humans can effectively find important information and ignore irrelevant parts in complex scenes, attention mechanisms have been introduced into DL for performance improvement (Niu et al., 2021). It is a way that dynamically adjusts the weights to different regions, thus enforcing the learning process to focus on important regions and obtain critical information from inputs (Wang et al., 2022b).

Various attention mechanisms have been proposed and applied to computer vision, which can be mainly categorized into spatial attention and channel attention according to the dimension operated. Spatial attention emphasizes interesting spatial domains, while channel attention considers the relations of feature maps between channels (Sdraka et al., 2022). The following lists the major studies that explored the integration of attention modules into meteorological downscaling networks.

- Convolutional block attention module (CBAM) (Woo et al., 2018) is a lightweight attention module combining channel attention module (CAB) and spatial attention blocks (SAB) in sequence. CAB uses max-pooling and average-pooling outputs with a shared network to enhance the attention to the key channels, while SAB applies average-pooling and max-pooling operations along the channel axis which are forwarded to a convolution layer, thus focusing on the inter-spatial relationship. CBAM was added to every convolutional layer of the proposed CNN for precipitation downscaling in Jin et al. (2022).
- Local attentive module (LAM) (Liu et al., 2019b) simultaneously uses spatial attention and channel attention to capture and aggregate global context and local texture of target local regions, which was designed in a SR network ABPN, the fundamental for the DeepCAMS downscaling method (Xiao et al., 2022).
- ResAttentionBlock (Tie et al., 2022) contains a channel attention branch and a spatial attention branch, and fuses the features after attention weighting. This attention unit was added at the end of the standard residual structure of the China Meteorological Administration's Land Data Assimilation System.
- Residual channel attention block (RCAB) (Zhang et al., 2018a) integrates channel attention into residual blocks in very deep residual channel attention networks (RCAN) for image SR, considering that high-frequency channel-wise features are informative. This attention block was used for wind downscaling (Liu et al., 2022).
- Attention gates (Oktay et al., 2018) that can filter the features propagated through the skip connections has been added to the upsampling step in the AU-Net for precipitation downscaling (Sun and Tang, 2020).
- Fused temporal cross attention mechanisms (FTCA) (Adewoyin et al., 2021) has been employed to improve the modeling ability for multiple spatio-temporal scales of weather dynamics in the TRU-NET downscaling model.
- Axial attention mechanism (Yu et al., 2022a) can consider both latitude and longitude relevance, and explore the long-range spatial similarity. It has been integrated into a terrain-guided flatten memory network (TIGAM) for downscaling complex meteorological elements (Yu et al., 2022b).

4. Downscaling-specific issues

Despite the similarities between meteorological downscaling and image SR, there are specific issues for DL-based meteorological downscaling to be thoroughly considered. This review addresses the major specific issues in recent studies as follows.

4.1. Downscaling factor

Image SR is an ill-posed problem that tries to obtain the HR images from the corresponding LR input. With the downscaling factor increasing, the recovery of the lost scene becomes more complicated, thus generally leading to the replication of improper information (Lepcha et al., 2023). Here downscaling factor refers to the magnification of resolution improvement through downscaling, similar to the scaling factor used in image super-resolution (Kim et al., 2016; Zhang et al., 2018b; Dai et al., 2019). Therefore, the scale factors of SR networks in the existing studies generally range from 2 to 8. However, the factor needed for the meteorological spatial downscaling to generate HR data is much higher, and has reached a maximal of 50x as shown in Table 1. The optimal choice of scaling factors and associated models over various conditions for downscaling tasks remain challenging (Dong et al., 2020).

To obtain higher downscaling factors, a common practice is to stack image SR networks, which may also allow for learning complex structures and features (Wang et al., 2021). For instance, Vandal et al. (2017) proposed a stacked downscaling network DeepSD based on the widely used image SR network SRCNN, and performed downscaling through multiple SR steps for precipitation. The original DeepSD is stacked by three SRCNNs (Dong et al., 2016a), the output of the former is used as the input of the latter, and each SRCNN performs the 2x downscaling. Each SRCNN applies the paired data of different resolutions obtained by bicubic downsampling on the same original dataset to perform separate training, which avoids the error propagation caused by the previous model. DeepSD finally downscaled precipitation data from 1° to 0.125°, which obtains the 8x downscaling results by stepping the process. Later, Kumar et al. (2021) duplicated the method for downscaling rainfall over India. In their work, two SRCNNs have been stacked for 4x downscaling at 0.25°, which verifies that DeepSD can outperform the conventional linear interpolation approaches. Passarella et al. (2022) adapted the FSRCNN, i.e. the resulting FSRCNN-ESM to reconstruct the Earth system model (ESM) simulations with a downscaling factor of 4 by using two stacked SRCNNs. Compared with the other SR approaches, the selected fundamental model FSRCNN has smaller convolution layers, which avoids over-smoothing and reduces computational costs (Dong et al., 2016b). The GAN-based PhIRE proposed by Stengel et al. (2020) also used two stacked SR networks with 10x and 5x respectively in two generators and achieved the wind field downscaling of 50x. The higher downscaling factor has also been realized by not only directly stacking the SR networks, but also stacking the basic down and up blocks (Liu et al., 2022), upsampling

blocks (Wang et al., 2021), deconvolution layers (Doury et al., 2022) etc.

As shown in Table 1, the current studies using GAN-based downscaling networks generally present higher downscaling factors (e.g., 50x for PhIRE (Stengel et al., 2020), 10x for DeepDT (Cheng et al., 2022) and VAE-GAN (Harris et al., 2022), 16x for SGAN (Leinonen et al., 2021) and GDRN (Liu et al., 2022)) with better identification of high-frequency details and data reconstruction quality by introducing perceptual loss. In addition, since the accuracy metrics in terms of root mean square error (RMSE), mean absolute error (MAE), etc. are considered as the major consideration for the meteorological data downscaling networks, in addition to the peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) widely adopted for image SR, the factor gained from the DL-based downscaling has not been restricted like the image SR task.

4.2. Spatial–temporal correlation

Different from natural images, meteorological data presents a more significant spatial and temporal correlation, and the spatial and temporal scale varies. The spatial–temporal correlation refers to the correlation among meteorological data in terms of space and time, encompassing the correlation between adjacent and long-range regions, and the correlation existing in time series data (Harilal et al., 2021; Rampal et al., 2022). Spatially, the meteorological state at a certain location is generally highly related to the corresponding neighbors (Yu et al., 2022b), and it is still considered a great challenge to capture the spatial variability of local and small-scale features such as extreme events (Hertig et al., 2019; Maraun et al., 2019; Wang et al., 2021). Except for the spatial correlation of adjacent regions, the meteorological variables can also exhibit long-term dependence among spatially non-contiguous or even far apart regions, which is referred to as teleconnection (Harilal et al., 2021). Besides, some phenomena are restricted to specific spatial locations. For example, typhoons occur in the ocean or coastal areas, which can be taken as prior information. Temporally, the present state of the system may influence the future states (Harilal et al., 2021), and the meteorological state can also present periodicity at different temporal scales (Yu et al., 2022b). Therefore, integrating spatial–temporal correlation information for downscaling is essential.

It has been widely accepted that DL can help identify detailed correlations and discover implied physical principles in Earth climate systems. The feature extraction enabled by CNNs has achieved impressive success in data modeling with spatial context, and CNN is the most widely employed architecture for spatial downscaling as shown in Section 3.1 and Table 1. Apart from the CNN-based methods, RNN and the recurrent structure have been considered as the major way to integrate the temporal correlation that exists in a time sequence of meteorological variables into downscaling networks (Zhang and Li, 2021).

Several studies improved RNNs for the spatial downscaling of meteorological data to improve performance. Harilal et al. (2021) presented a recurrent convolutional LSTM-based statistical downscaling approach from coarse-resolution predictors (e.g., pressure fields) to fine-resolution variables (e.g., temperature and precipitation), which helps to account for complex and non-linear spatial and temporal dependencies. As shown in Fig. 5, this method uses stacked ConvLSTM blocks to encode the temporal information in the input variable sequence. Chou et al. (2021) developed a SR-based convolutional LSTM network to capture both spatial and temporal variability, and tested its robustness and predictability over monthly precipitation data in China. The proposed model adopts a stacked ConvLSTM architecture which consists of three stacked ConvLSTM blocks and a SR block, which help to utilize the temporal dependence and retain the spatial correlation. Similarly, Anh et al. (2019) proposed to use the LSTM network to downscale precipitation for the Vietnamese Mekong Delta. Furthermore, Zhang and Li (2021) considered efficiently capturing the nonlinear and temporal forward–backward correlations using past and future features, and proposed a downscaling method for wind energy based on BiGRU.

Besides, there are endeavors to integrate recurrent structures in downscaling networks. Adewoyin et al. (2021) presented TRU-NET, a temporal recurrent UNet model considering the multi-scale spatial and temporal behavior of weather processes. The structure designed in the TRU-NET to model the spatial–temporal information mainly contains a stack of bi-directional ConvGRU layers with a novel fused temporal cross-attention mechanism FTCA. Leinonen et al. (2021) downscaled the time-evolving atmospheric fields with a GAN-based stochastic SR network. Unlike most GANs, in addition to the convolutional and residual blocks to learn the spatial structure, recurrent layers in the form of ConvGRUs have been added to the generator to learn the temporal evolution.

Instead of using temporal information for spatial downscaling, Xiao et al. (2022) considered the spatial and temporal dimensions separately and developed a DL framework for spatial–temporal downscaling DeepCAMS. It can maintain the spatial consistency and temporal continuity, and has been used to generate a long-term (2003–2020) high spatio-temporal resolution global PM2.5 dataset. DeepCAMS handles the temporal and spatial dimensions separately, and the temporal downscaling step performed before spatial downscaling has been motivated by the video frame interpolation network, which can explore the link between missing information and existing estimates.

In particular, a rather different solution used in the HR precipitation nowcasting might inspire the downscaling task. Ravuri et al. (2021) applied the conditional GAN and used two loss functions in model training, i.e. a separate spatial and temporal discriminator. The temporal discriminator has been designed to impose consistency in temporal sequences. This approach has not yet been explored for the downscaling task, but might give some hints to better utilize the temporal sequences for downscaling performance improvement.

4.3. Auxiliary variables

Different from single image SR in computer vision, which tries to reconstruct the HR image with enhanced details from a corresponding single degraded LR image, the spatial downscaling of meteorological data should capture the physical and dynamical processes present in the atmosphere and the complex interactions between components of Earth System (Doury et al., 2022; Rampal et al., 2022). Hence, the relationship among various meteorological variables is complicated, and they have great influences on each other. For example, the wind is closely related to the temperature and the air mass, which are the key impact factors of pressure difference, and moreover, the humidity influences air mass (Yu et al., 2022b). Selecting appropriate auxiliary variables for downscaling is crucial to simulate the relationship between predictors and predictands accurately. The auxiliary variables, generally a subset of predictors, refer to the meteorological variables used to assist the downscaling of target variables (Harris et al., 2022; Quesada-Chacon et al., 2022; Yu et al., 2022b). Note that these predictors might be called downscaling factors in the studies for downscaling other satellite datasets (Sun and Cui, 2021). Currently, there is no uniform standard for auxiliary variables used in the DL-based spatial downscaling, even for the same target predictand. The auxiliary variables vary greatly due to the study area, data acquisition time, data availability, etc.

Nonetheless, some studies simplified the above-mentioned impacts, and still considered the spatial downscaling as a single image SR task. Therefore, the corresponding designed downscaling network handles the mapping from a single LR variable to its corresponding HR downscaled result. For instance, Wang et al. (2021) downscaled daily precipitation and temperature with SRDRN using the highly noisy simulation as the single predictor, and can achieve notable improvement compared to traditional statistical downscaling methods. However, they have considered adding physical relevant variables from GCMs,

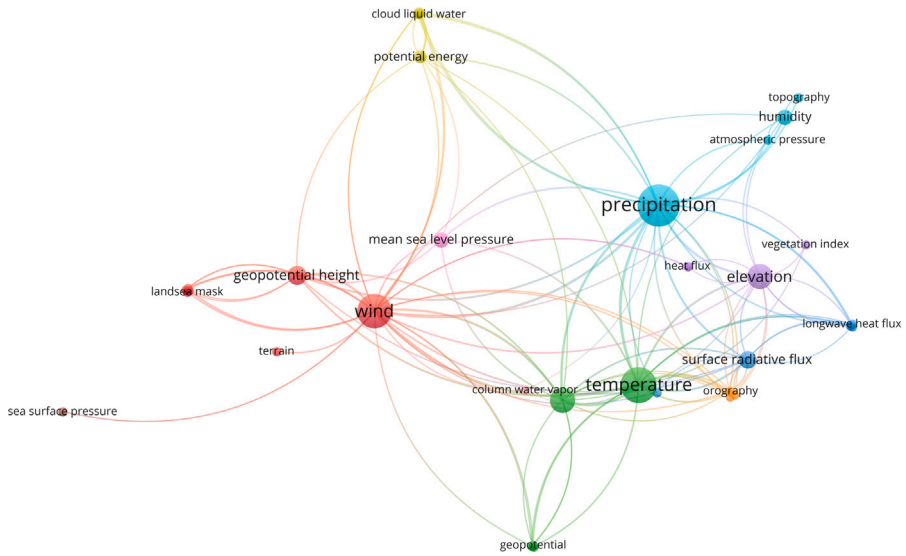


Fig. 7. Network analysis on the terms contained in the auxiliary variables of the reviewed studies in Table 2. The frequency is represented by the size, while the color represents statistically derived clusters. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

e.g. temperature, humidity, and geopotential height at different vertical levels as input channels for future work. To downscale wind, Stengel et al. (2020) proposed an adversarial SR method based on the SR network SRGAN, and adjusted the network input as two channels with zonal and meridional wind components. Similarly, in a previous work to downscale satellite sea surface wind based on GAN with dual learning, the input channels include zonal, meridional wind components and an additional mask, and have achieved superior performance compared with representative downscaling methods (Liu et al., 2022).

To depict the multi-variable coupling in the downscaling process for better performance (Xiao et al., 2022), a common practice is to impose additional meteorological variables as input, i.e. predictors (Yu et al., 2022b). Table 2 summarizes the predictors that include auxiliary variables used in the existing work of DL-based spatial downscaling for different predictands. A network analysis on terms contained in auxiliary variables of the review studies in Table 2 was performed using the VOSviewer software, and the corresponding results are shown in Fig. 7. The major clusters are precipitation, temperature and wind, the target downscaled variables. Fig. 7 also helps to identify the interrelationships between distinct variables, as well as the closely related auxiliary variables to the clusters clearly from the existing literatures.

Several studies accounted for the interactions between the states of atmosphere and topography, and introduced the topography related data such as the digital elevation model (DEM) as the only additional inputs. For example, for precipitation downscaling, DeepSD (Vandal et al., 2017) and Nest-UNet (Sha et al., 2020b) both used the static topographical features elevation as an auxiliary predictor, especially to generate more accurate patterns in complex terrain. Similarly, the elevation data have been used to provide more detailed terrain semantics for the temperature spatial downscaling in CLDASSD (Tie et al., 2022) and UNet-AE (Sha et al., 2020a). ResLap selected 3D precipitation, 3D topography, and 3D humidity as available variables, and used a convolutional layer to extract shallow hierarchical features from the multi-channel images as the final inputs of the precipitation downscaling model (Cheng et al., 2020). Later, DeepDT added 3D temperature in addition to 3D precipitation, 3D topography, and 3D humidity (Cheng et al., 2022).

Many studies tend to use a more comprehensive set of predictors, referring to traditional statistical downscaling methods. Motivated by the ecPoint model (Hewson and Pillosu, 2021) and domain knowledge, Harris et al. (2022) used 9 fields from the ECMWF's Integrated Forecast System (IFS) model, including total precipitation, convective precipitation, surface pressure, top-of-atmosphere (TOA) incident solar radiation, convective available potential energy, total column cloud liquid water, total column water vapor, U & V wind at 700 hPa. Several researches used 20 standard predictors proposed in the VALUE framework, including the geopotential height, zonal and meridional wind, temperature, and specific humidity at four different vertical levels (1000, 850, 700 and 500 hPa) (Bano-Medina et al., 2020, 2021; Sun and Lan, 2021). Unlike the above work, although (Jin et al., 2022) used a variety of variables (e.g. seasonal anomaly field of zonal wind speed at 200 hPa, geopotential height at 500 hPa, specific humidity, zonal and meridional wind speeds at 500, 700 and 850 hPa, mean sea level pressure, and 2-m temperature) as the inputs, they tried to feed the CNNs sufficient raw grid values of general circulation and extract useful features automatically, rather than selecting hand-crafted features from experience as predictors. Furthermore, Rampal et al. (2022) listed a full set of variables (convective available potential energy, mean sea level pressure, specific humidity at 850 and 500 hPa, air temperature 2 m and 850 hPa, zonal wind, meridional wind, vertical wind at 850 hPa, geopotential height at 850 hPa, total precipitation), assessed the importance of different subsets of predictor variables, and tried to interpret the impact of a particular variable on the downscaling network.

In general, the predictors with additional variables are stacked as multi-channel inputs (Pan et al., 2019; Bano-Medina et al., 2021; Jin et al., 2022; Adewoyin et al., 2021). However, different variables vary in data distribution, and how to integrate these predictors to improve the downscaling performance remains challenging. In particular, in the UNet-AE for temperature downscaling, Sha et al. (2020a) modified the original UNet, and assigned an additional output branch and loss function to handle the elevation information. In the TIGAM network for wind downscaling, Yu et al. (2022b) also adopted two input branches that correspond to wind and other variables, and employed an auxiliary terrain-guided loss to impose the restriction between terrain and wind distribution in addition to the reconstruction loss. Besides, similar to image SR, the multi-task learning strategy can be introduced to learn the relationship and capture the domain-specific knowledge among the spatial downscaling process for different but related predictands, thus promoting downscaling performance and generalization ability (Yan and Chang, 2021).

4.4. LR-HR paired dataset

Various benchmark datasets have been constructed with pairs of LR and HR images for image SR. These images are generally collected in a

Table 2
A summary of auxiliary variables for the DL-based meteorological data downscaling, where predictors show the meteorological variables that are used to implement the downscaling for predictands, i.e. the target variables. The predictors include auxiliary variables.

Predictands	Predictors	Method
Precipitation	Total precipitation, convective precipitation, surface pressure, TOA incident solar radiation, convective available potential energy, total column cloud liquid water, total column water vapor, U & V wind at 700 hPa.	GAN, VAE-GAN (Harris et al., 2022)
	Seasonal anomaly field of zonal wind speed at 200 hPa, geopotential height at 500 hPa, specific humidity, zonal and meridional wind speeds at 500, 700 and 850 hPa, mean sea level pressure, and 2-m temperature.	Customized CNN (Jin et al., 2022)
	Convective available potential energy, mean sea level pressure, specific humidity at 850, 500 hPa, air temperature 2 m, 850 hPa, zonal wind & meridional wind & vertical wind at 850 hPa, geopotential height at 850 hPa, total precipitation.	Linear CNN, Non-linear CNN (Gamma) (Rampal et al., 2022)
	Zonal and meridional wind, temperature, geopotential and specific humidity at the 1000, 850, 700 and 500 hPa.	Modified climate4R (Quesada-Chacon et al., 2022)
	ERA5 precipitation, high-resolution simulated precipitation, elevation data from shuttle radar topography mission.	SPCN (Jiang et al., 2021)
	Precipitation, elevation, humidity, atmospheric pressure, and wind-velocity components in the three directions.	Augmented-ConvLSTM (Harilal et al., 2021)
	3D precipitation, 3D topography, 3D temperature, and 3D humidity.	DeepDT (Cheng et al., 2022)
	3D precipitation, 3D topography, and 3D humidity.	ResLap (Cheng et al., 2020)
	Zonal and meridional wind at 850 hPa, specific humidity, geopotential at 500 hpa, air temperature, and total column water vapor in the entire vertical column.	TRU-NET (Adewoyin et al., 2021)
	Precipitation, and elevation.	Nest-UNet (Sha et al., 2020b), DeepSD (Vandal et al., 2017)
	Precipitation, elevation, vegetation index, and air temperature.	AU-Net (Sun and Tang, 2020)
	Precipitation, minimum temperature, maximum temperature, specific humidity, mean sea level pressure, zonal and meridional wind.	Convolutional LSTM (Chou et al., 2021)
	Geopotential height at different pressure levels, total column precipitable water.	CNNs (Pan et al., 2019)
Wind	Analysis data: wind components U & V, perpendicular deviation of wind vector, atmospheric stability, ground surface sensible heat flux, elevation of the model's terrain; high-resolution topographic descriptors: elevation, slope and aspect.	Wind-Topo (Dujardin and Lehning, 2022)
	Wind fields, terrain.	TIGAM (Yu et al., 2022b)
	Sea surface wind speed at 10 m, wind speed at 850 hPa, sea surface pressure.	BiGRU (Zhang and Li, 2021)
	Meridional wind, zonal wind, land surface elevation, land–sea mask, boundary layer height, forecast surface roughness, geopotential height at 500 hPa.	DeepRU (Höhlein et al., 2020)
Temperature	Temperature, digital elevation model.	CLDASSD (Tie et al., 2022)
	2D variables: Geopotential & specific humidity & temperature at 850, 700, 500 hPa, zonal and meridional wind at 850, 700, 500 hPa + surface, sea level pressure, total aerosol optical depth forcing; 1-D variables: daily spatial means of 2D variables, daily spatial standard deviation of 2D variables, total anthropogenic greenhouses gas forcings, solar and ozone forcings, seasonal indicators.	RCM-Emulator (Doury et al., 2022)
	Elevation, maximum/minimum 2-m temperature.	UNet-AE (Sha et al., 2020a)
Multiple variables: precipitation, temperature, (surface radiative fluxes (Passarella et al., 2022))	Surface: maximum temperature, mean temperature, U10 & V10, accumulated precipitation; upper level at 850, 700, 500 hPa: specific humidity, temperature, U, V; invariant variable at surface: angle of sub-grid-scale orography, anisotropy of sub-grid-scale orography, standard deviation of filtered subgrid orography, standard deviation of orography, geopotential, latitude, longitude; temporal: day of year; elevation, elevation difference, topographic position index.	convCNPs (Vaughan et al., 2022)
	Geopotential, air temperature, specific humidity, zonal wind, meridional wind (all variables at 500, 700, 850 and 1000 hPa).	CNN1, CNN10 (Bano-Medina et al., 2021)
	Surface temperature, shortwave heat flux, longwave heat flux, precipitation convective rate, large scale precipitation rate, elevation.	FSRCNN-ESM (Passarella et al., 2022)
	Geopotential height, zonal and meridional wind, temperature, and specific humidity at four vertical levels (1000, 850, 700 and 500 hPa).	CNNs (Bano-Medina et al., 2020; Sun and Lan, 2021)

wide variety of scenarios. In particular, the LR images can be obtained either by performing bicubic interpolation on the HR images (e.g., Urban100), or with different degradation kernels (e.g., DIV2K). Some datasets reflect the real image degradation process (e.g., DRealSR), which alleviates the limitation of traditional analog image degradation. By contrast, the LR-HR paired datasets for the spatial downscaling of meteorological data are hardly publicly available. Currently, there are two types of composition for the paired datasets for the spatial downscaling model training. One is to obtain LR data from HR data by downsampling to form paired data, while the other is by finding spatio-temporally matched HR and LR data products, which can come from different data sources.

Additionally, when it comes to downscaling, it is important to carefully consider how to partition the dataset. The image SR task typically divides datasets into training, validation and testing sets randomly. Since the meteorological data exhibit temporal correlations over different time scales, e.g. from up to a few days to seasonal and interannual variations, the datasets are usually divided into consecutive months or years to avoid overfitting (Höhlein et al., 2020). For instance, Harris trained the precipitation downscaling model with data from 2016 to 2018, and used the data from 2019 as the validation set for hyperparameter tuning and model performance evaluation, and held out the data from 2020 for the final testing (Harris et al., 2022). Jin et al. (2022) selected the data from 1994 to 2008 for model training, the data from 2009 to 2012 for validation, and the data from 2013 to 2016 for testing. Various works, for example, Rampal et al. (2022), Passarella et al. (2022) and Liu et al. (2022), have performed similar dataset division strategies according to time periods.

5. Datasets, toolsets and metrics

5.1. Datasets and variables

This subsection presents the status of meteorological dataset for the DL-based downscaling and the target variables, i.e. predictands in the existing DL-based downscaling studies, e.g. temperature, precipitation, wind. It may be of interest to the data demands for the community. Different from the image SR task in which benchmark datasets (such as DIV2K, Flickr2K and Set5, etc.) are available for training, validating, testing and comparing the capability of methods, there are few benchmark datasets for the spatial downscaling task of meteorological data. Therefore, we give a summary of the LR and HR dataset of the DL-based meteorological data downscaling methods in Table 3. It attempts to facilitate the development, validation and comparison of relevant DL methods.

The meteorological data typically has a spatial resolution of more than ten kilometers, with a time resolution ranging from hours to days. The dynamical methods are used to achieve downscaling at a factor less than the DL-based methods, with significant computational expense. Meanwhile, the variables involved in the traditional statistical downscaling methods and DL-based methods are fairly similar. The generated downscaled products at sub-kilometer resolution have been applied in the refined researches for the prediction or evaluation of climate change (Dai et al., 2020), meteorological resources (Stengel et al., 2020), and extreme events (Shi, 2020).

Most DL-based downscaling studies focus on meteorological variables, e.g. precipitation, temperature, and wind. These LR and HR datasets are obtained mainly from observations, simulations of GCMs, and reanalysis data, released by institutions shown in Table 3. Among the listed datasets, some of the LR and HR datasets are both accessible, while parts of the LR datasets are synthesized by upscaling the HR datasets, such as PRISM and gridMet. The datasets will be preprocessed by gridding and padding, before they are fed into the DL networks. In addition to the major variable studied, some auxiliary variables are also fed into the networks as shown in Table 2, hence the inputs of DL-based downscaling methods are generally multi-channel. The following demonstrates the status of major DL-based downscaled variables, including precipitation, temperature, and wind.

HR precipitation products play an important role in climate change assessment, crop yield management, hydrological forecasting, surface and groundwater resource planning, etc. Currently, the resolution of precipitation products is generally coarser than 0.25°. For example, the Climate Research Unit (CRU) provides a suite of climate variables, including precipitation at a resolution of 0.25° (Harris et al., 2014). The Global Precipitation Climatology Center (GPCC) is able to collect, perform quality control, and analyze rain gauge data from across the globe, and a database has been constructed with four globally gridded monthly precipitation products, i.e. the Climatology (CLIM) V2011, the Full Data Reanalysis (FD) V6, the Monitoring Product (MP) V4, and the First Guess Product (FG), the resolutions of which vary from 0.25° to 2.5° (Becker et al., 2013). Sensors onboard satellites provide access to global precipitation measurements conveniently, while satellite precipitation products include Climate Prediction Center morphing technique (CMORPH) (Joyce et al., 2004), Tropical Rainfall Measuring Mission (TRMM), Multisatellite Precipitation Analysis (TMPA) (Huffman et al., 2007), Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks (PERSIANN) (Hong et al., 2004), etc, with resolutions varying from 0.04° to 2.5°.

Temperature is an important meteorological factor affecting global climate change, and plays an important role in biogeochemical and hydrosphere cycles (Han et al., 2020). The focus of studies around temperature downscaling varies slightly, for instance, sea surface temperature (Hoffmann et al., 2016), daily maximum and minimum temperature (Wang et al., 2021; Sha et al., 2020a). Most of temperature data were obtained from the atmospheric assimilation dataset and the surface gridded fusion dataset, such as gridMET (Abatzoglou, 2013), E-OBS (Cornes et al., 2018), CFDT (Chen et al., 2011) and SMBFD (Han et al., 2019b). Their resolutions reach a kilometer level, higher than satellite observations, and simulations of GCMs, and generally taking as HR datasets. Some of the LR datasets are from reanalysis systems. The maximum downscaling scale for temperature is up to 24.

The current downscaling studies for wind fields mainly focus on HR wind energy resources (Stengel et al., 2020), near-surface wind in complex terrain Dujardin and Lehning (2022), and sea surface wind (Liu et al., 2022). The most widely used datasets are mainly from reanalysis systems, satellite observations and even GCMs products, e.g. ERA5 (Hersbach et al., 2020), WindSat Wentz et al. (2013), NCAR CCSM4 (Taylor et al., 2012). The maximum downscaling scale reaches 50, producing HR wind data at a resolution of 2 km.

5.2. Available toolsets

In view of some tedious tasks in the downscaling of meteorological data, such as data acquisition, processing, and selection of comparison methods, Table 4 shows a set of existing available downscaling tools with brief descriptions. Part of the tools have been introduced in the previous sections (i.e. tools from DeepSD to VAE-GAN), and the tools from SDSM to ClimaDown are based on classical statistical downscaling methods, while the rest of downscaling tools that integrate multiple downscaling models or build generic models will be described in detail as follows.

For transparency and reproducibility in downscaling research, DeepDownscaling (Bano-Medina, 2020) hosts and maintains six implementations of the papers by the Santander MetGroup for statistical downscaling based on DL, i.e. 2018_Bano_CI (Chen et al., 2018), 2019_Bano_CI (Brajard et al., 2019), 2020_Bano_GMD (Bano-Medina et al., 2020), 2020_Bano_CI (Bano-Medina, 2020), 2020_Bano_CD (Bano-Medina et al., 2021), 2022_Bano_GMD (Baño-Medina et al., 2022). These implementations are based on climate4R, a set of R packages for open climate data access, post-processing, visualization, and model validation (Iturbide et al., 2019).

Table 3
A summary of the meteorological data in LR and HR for downscaling in the literatures reviewed. It shows the training datasets with citations used in downscaling methods, which are classified by the predictands. The spatial resolutions of datasets are also provided.

Predictands	LR dataset	Spatial resolution	HR dataset	Spatial resolution	Method
Precipitation	PRISM (Daly et al., 2008)	1°	PRISM	0.125°	DeepSD (Vandal et al., 2017)
	GPCC (Schröder et al., 2019)	2.5°	GPCC	0.5°	CDN (Mu et al., 2020b)
	ECMWF ERA5 (Hersbach et al., 2020)	31 km	VCSN (Tait et al., 2006)	0.05°	Non-Linear CNN (Rampal et al., 2022)
	PRISM (downsampled) (PRISM Climate Group, 2004)	0.25°	PRISM (PRISM Climate Group, 2004)	4 km	Nest-UNet (Sha et al., 2020b)
	TRMM (Huffman et al., 2007)	0.25°	PRISM (Daly et al., 2008)	1 km	AU-Net (Sun and Tang, 2020)
	ERA5 (Hersbach et al., 2020)	0.25°	ReKIS (ReKIS, 2021)	1 km	Modified climate4R (Quesada-Chacon et al., 2022)
	IMD (downsamped) (Rajeevan et al., 2008)	1°	IMD (Rajeevan et al., 2008)	0.25°	Stacked SRCNNs (Kumar et al., 2021)
	NCAR CESM1 CAM5 (Lauritzen et al., 2018)	1.15°	IMD4 (Pai et al., 2014)	0.25°	Augmented-ConvLSTM (Harilal et al., 2021)
	ECMWF ERA5 (downsampled) (Hersbach et al., 2020)	65 km	E-OBS (Cornes et al., 2018)	8.5 km	TRU-NET (Adewoyin et al., 2021)
	gridMET (Abatzoglou, 2013)	1°	gridMET	4 km	SRDRN (Wang et al., 2021)
	HRSP3 (Zhou et al., 2018)	0.26664°	HRSP3	0.03333°	SPCN (Jiang et al., 2021)
	NCAR CCSM4 (Taylor et al., 2012)	1.25° × 0.9°	ESSD (Peng et al., 2019)	0.25°	ConvLSTM (Chou et al., 2021)
	MCH-RZC (Germann et al., 2015)	1/16 km	MCH-RZC	1 km	SGAN (Leinonen et al., 2021)
Temperature	gridMET	1°	gridMET	4 km	SRDRN
	ERA-Interim (Dee et al., 2011)	2°	E-OBS (Cornes et al., 2018)	0.05°	CNN1, CNN10 (Bano-Medina, 2020)
	ERA-Interim (Dee et al., 2011)	2°	CFDT (Chen et al., 2011)	0.1°	CNN1, CNN10 (Sun and Lan, 2021)
	PRISM (downsampled) (Strachan and Daly, 2017)	0.25°	PRISM (Strachan and Daly, 2017)	4 km	UNet-AE (Sha et al., 2020a)
	ESGF (downsampled) (Cinquini et al., 2014)	1°	ESGF (Cinquini et al., 2014)	0.25°	FSRCNN-ESM (Passarella et al., 2022)
	CLDAS (Han et al., 2020)	0.0625°	SMBFD (Han et al., 2019b)	0.01°	CLASSD (Tie et al., 2022)
Wind	ECMWF ERA5 (Hersbach et al., 2020)	31 km	ECMWF HRES (ECMWF, 2019)	9 km	DeepRU (Höhlein et al., 2020)
	ERA-Interim (Dee et al., 2011)	1.5°	ECMWF ERA5 (Hoffmann et al., 2019)	0.25°	BiGRU (Zhang and Li, 2021)
	NCAR CCSM4 (Taylor et al., 2012)	0.9° × 1.25°	NREL WIND Toolkit (Draxl et al., 2015)	2 km	PhIRE (Stengel et al., 2020)
	WindSat Wentz et al. (2013)	0.25°	–	–	GDRN (Liu et al., 2022)

Table 4
A summary of downscaling toolsets. It gives a set of open-access downscaling tools with the basic descriptions and access URLs.

Tool	Features	URL
DeepSD (Vandal et al., 2017)	Data download and process (designed for 8 × precipitation)	https://github.com/tjvandal/deepsd
convCNPClimate (Vaughan et al., 2022)	Designed for temperature, precipitation data from sites	https://github.com/anna-184702/convCNPClimate
RCM-Emulator (Doury et al., 2022)	Designed for 12.5 × temperature	https://github.com/antoinedoury/RCM-Emulator
modified climate4R (Quesada-Chacon et al., 2022)	Based on climate4R, training different models by changing the parameters (for 25 × Precipitation)	https://github.com/dquesadacr/Rep_SDDL
TRU-NET (Adewoyin et al., 2021)	For 7 × precipitation prediction in 28 days	https://github.com/Akanni96/TRUNET
Augmented-ConvLSTM (Harilal et al., 2021)	Multiple variables inputs (for 4.6 × temperature and precipitation)	https://github.com/cryptonymous9/Augmented-ConvLSTM
PhIRE (Stengel et al., 2020)	Contains two model for 5 × and 10 × downscaling, (for wind, solar)	https://github.com/NREL/PhIRE
Drnngan (Leinonen et al., 2021)	For 16 × precipitation and cloud optical thickness downscaling series	https://github.com/jleinonen/downscaling-rnn-gan
VAE-GAN (Harris et al., 2022)	For 10 × precipitation downscaling with visualization	https://github.com/ljharris23/public-downscaling-cgan
SDSM	Widely used Statistical Downscaling model implemented by R language	https://github.com/leerichardson/sdsmR
CDFt (Michelangeli et al., 2009)	A probabilistic downscaling method based on cumulative distribution functions transformation	https://cran.r-project.org/web/packages/CDFt/index.html
RainFARM (Rebora et al., 2006)	A metagaussian stochastic spatio-temporal precipitation downscaling method based on fourier power spectrum	https://cran.r-project.org/web/packages/rainfarmr/index.html
spdownscale	Spatial downscaling for GCMs/RCMs using quantile–quantile bias correction	https://cran.r-project.org/web/packages/spdownscale/index.html
BCSD (Wood et al., 2004)	A simple and effective downscaling method based on bias correction spatial disaggregation	https://github.com/tjvandal/bcsd-python
ClimDown	For statistical downscaling of coarse scale GCM output, needs for high computing resources	https://github.com/pacificclimate/ClimDown
DeepDownscaling (Bano-Medina, 2020)	Contains six implementations based on climate4R	https://github.com/SantanderMetGroup/DeepDownscaling
DL4DS (Gomez Gonzalez, 2023)	Provides supervised and unsupervised methods with custom scale factor	https://github.com/carlos-gg/dl4ds
ClimateLearn	Provides access to multiple datasets and methods	https://github.com/aditya-grover/climate-learn
ClimaX (Nguyen et al., 2023)	General model based on ViT for forecasting and downscaling tasks	https://microsoft.github.io/ClimaX/
VALUE (Maraun et al., 2019)	An open European framework to validate and compare downscaling methods	http://www.value-cost.eu/

Deep learning for empirical downscaling (DL4DS) is a released Python package that implements a wide variety of DL networks for downscaling gridded Earth science data (Gomez Gonzalez, 2023). The package can be simply installed by the pip installation tool, and it provides both supervised and unsupervised downscaling methods, depending on whether the conditional adversarial module is applied. These implementations consist of different backbones and upsampling methods, which form various downscaling methods that allow for custom scale factors and auxiliary variables. In addition, the package offers basic dataset loading, generating, and processing functions.

ClimateLearn is a Python package, similar to DL4DS. It provides access to two types of datasets, ECMWF ERA5 (Hersbach et al., 2020) and WeatherBench ERA5 (Rasp et al., 2020). The baseline models cover several DL methods, such as residual convolutional neural networks, U-nets, and vision transformers. ClimateLearn has been applied to weather forecasting and downscaling for various variables, as well as generating visualizations for these applications.

ClimaX is a general foundation model, dedicated to being simple and flexible to use for different studies of weather and climate science (Nguyen et al., 2023). The models based on standard Vision Transformer (ViT), can be applied to various tasks of forecasting (i.e. global forecasting, regional forecasting, and Sub-seasonal to seasonal cumulative prediction), climate projection, and downscaling. There are different models trained for each task by data from the Coupled Model Intercomparison Project (CMIP) (Zhou et al., 2014) and ERA5. Moreover, ClimaX supports efficient scalable distributed training, implemented by PyTorch Lightning.

In addition, given the differences in research targets, contents, regions, and evaluations in downscaling studies, it is difficult to make a multifaceted visual comparison between different methods. Therefore, an open European network for validating and integrating downscaling methods, VALUE (Maraun et al., 2019) is built, which provides a framework for users to validate and compare their methods conveniently under uniform conditions. The framework can support various experiments with different observational, pseudo-realistic reference data, and validate the methods from all sides via a validation tree.

5.3. Evaluation metrics

Evaluation metrics are crucial in assessing the performance and effectiveness. They provide objective measures to quantify and compare the quality, accuracy, or efficiency of different approaches. For the evaluation of meteorological data spatial downscaling, the metrics commonly used can be divided into two categories.

Table 5
The most popular evaluation metrics for downscaling methods assessment.

Metric	Range	Description
B	$(-\infty, +\infty)$	The average error between the predictions and the actual values.
SD	$[0, +\infty)$	Quantifies the amount of variation or dispersion in a set of data values.
MAE	$[0, +\infty)$	The average absolute error between the predicted and actual values, indicates the actual error intuitively.
MSE	$[0, +\infty)$	The average absolute error between the predicted and actual values. It is particularly useful for penalizing large errors.
RMSE	$[0, +\infty)$	The square root of the average squared error between the predicted and actual values, sensitive to abnormal value
PSNR	$[0, +\infty)$	The ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation. The higher the PSNR, the higher the quality of the image or signal.
R	$[-1, 1]$	The strength and direction of the linear relationship between predicted and actual values. The value close to 1 indicates a strong positive correlation, close to −1 indicates a strong negative correlation, and close to 0 indicates no correlation.
R^2	$[0, 1]$	Measures the ability of the model to explain the variability of the target variable. $R^2 = 1$ indicates that the model is able to account for all the variability of the target variables, i.e., the model's predictions are completely accurate. $R^2 = 0$ means that the model cannot account for the variability of any of the target variables, i.e., the model's predictions are no better than using the average of the target variables directly.
SSIM	$[-1, 1]$	Measures the similarity between two images. A value close to 1 indicates that the two images are very similar. It considers changes in structural information, luminance, and contrast, which aligns better with human visual perception.
KGE (Vrugt and de Oliveira, 2022)	$(-\infty, 1]$	Considers the mean, variance, and correlation of model predictions. The closer the KGE value is to 1, the better the model performance. It is derived from Nash–Sutcliffe Efficiency (NSE), less sensitive to extreme events.
EV (LaHuis et al., 2014)	$(-\infty, 1]$	Measures the ability of a model to explain the variance in the data. The closer the value is to 1, the better the explanatory power of the model.
ROCSS (Manzanas et al., 2014)	$[-1, 1]$	A comprehensive measure of the performance of a binary classifier as its discrimination threshold is varied. It is derived from the receiver operating characteristic curve. A value of 1 indicates perfect classification, a value of 0 indicates performance equivalent to random guessing, and a value of −1 indicates completely incorrect classification.
P98 (Bano-Medina et al., 2021)	$(-\infty, +\infty)$	A statistical measure that represents that 98 percent of the data points are below this value in all data. It can be used to understand and interpret the distribution of data, especially in understanding the tail behavior of the distribution.
CRPS (Astuti, 2019)	$[0, +\infty)$	Evaluate the accuracy of a predicted probability distribution in comparison to an observed probability distribution. A CRPS of 0 indicates a perfect match between the predicted and observed distributions. Larger values of CRPS indicate greater discrepancy between the predicted and observed distributions.
FSS (Skok and Roberts, 2018)	$[0, 1]$	Assess how well the forecasted and observed fields agree in terms of the spatial distribution of a particular variable. An FSS of 1 indicates a perfect match between the forecast and the observed field. An FSS of 0 indicates no spatial overlap at all between the forecast and the observed field.

The first category is error metrics which measure the difference between the predicted and actual values. This set of metrics mainly includes bias (B), standard deviation (SD), mean absolute error (MAE), mean squared error (MSE), root mean square error (RMSE), and peak signal-to-noise ratio (PSNR). They primarily assess the discrepancy between predicted and actual values, and can also be utilized as loss functions for model optimization.

The other is model performance evaluation metrics that evaluate the model from special aspects. It contains metrics from different aspects, including correlation coefficient (R), coefficients of determination (R^2), structural similarity (SSIM), Kling–Gupta efficiency (KGE) (Vrugt and de Oliveira, 2022), explained variance (EV) (LaHuis et al., 2014), relative operating characteristic skill score (ROCSS) (Manzanas et al., 2014), 98th percentile (P98) (Bano-Medina et al., 2021), continuous ranked probability score (CRPS) (Astuti, 2019), fractions skill score (FSS) (Skok and Roberts, 2018). They evaluate the model based on its goodness of fit, consistency, generalization, and predictive capability for extreme values.

Table 5 summarizes these evaluation metrics above and provides their brief introductions. In most downscaling research, it is customary to select two or more different types of evaluation metrics to holistically evaluate the model's performance, and the most widely used combinations in current research are RMSE, MAE, and R^2.

6. Challenges and future perspectives

Although significant progress has been made in improving the resolution, accuracy, and efficiency of downscaling, there are still several important issues that remain to be addressed. It can be commenced from several aspects, including unsupervised downscaling to adapt the scenarios lacking HR ground truth, improving the model generalization performance across diverse contexts (e.g., geographical regions, sensing platforms, temporal periods), downscaling uncertainty quantification, and physical-informed downscaling to enhance interpretability of results.

6.1. Unsupervised downscaling

Most existing downscaling methods are supervised and rely on paired data with low and high spatial resolution. However, obtaining spatio-temporal matching LR-HR data in real scenarios is challenging. As a result, generating HR results with scarce-paired samples or even only LR data directly has become a major concern for future studies.

The SR networks on natural images generally use external datasets to train the models to restore the local features of images, however, ignore the feature similarities within a single image. Therefore, there

are networks, such as RCAN (Zhang et al., 2018a) and SAN (Dai et al., 2019) introducing the non-local attention mechanism to extract the self-similarity of internal features of images. Since these networks only consider the features at the same scale, and ignore the hidden information between features of different scales, a cross-scale and non-local attention network (CSNLN) (Mei et al., 2020) was then proposed. CSNLN contains a self-exemplar mining (SEM) block that can facilitate prior information extraction and learning by using multi-branch structures.

These methods bring ideas for unsupervised learning in image SR. Several exploratory studies, e.g. ZSSR (Shocher et al., 2018) and MZSR (Soh et al., 2020), provide preliminary solutions for unsupervised training through learning the internal features of images and leveraging large amounts of external data for transfer learning. Moreover, there are some other unsupervised learning based methods, e.g. cycle-CNN (Wang et al., 2019b), HISR-GAN (Zhang et al., 2021b). Cycle-CNN uses two generative CNNs for SR and down-sampling, forming an inverse auto-encoder construct. HISR-GAN uses two generators to generate LR images from HR images and SR images from LR images, similar to the auto-encoder structure. Therefore, unsupervised downscaling can also adopt the ideas in these methods.

For instance, a preliminary work (Liu et al., 2022) that combines GAN and dual learning (Liu et al., 2022) has been presented to address this issue. The proposed method is able to generate data with higher resolution than the original data without a pre-prepared paired LR-HR dataset. The network obtains synthetic LR data during training by randomly selecting and performing a downsampling method on the origin data from the set of commonly used nearest neighbor, bilinear, and bicubic interpolations. Generative adversarial networks are designed to learn the features and generate data with the same distribution as original data. Moreover, the method includes a dual regression network to learn the mapping from origin to LR data, and the dual-learning networks are used for both supervised and unsupervised learning. This approach provides a solution for downscaling without paired data, and experimental results show better downscaling accuracy and reconstruction quality with more fine-grained details.

6.2. Transformer for downscaling

As shown in Table 1, CNNs are considered a fundamental component in downscaling, while RNNs and GANs also contribute a lot. Due to the receptive field limitation of convolutional kernels, the restriction of the contextual information processed and long-range scene perception may lead to the performance degradation (Luo et al., 2016). Subsequently, researchers started to incorporate a self-attention mechanism into CNNs to model long-range relationships. Recently, transformer, a new type of neural network in an encoder–decoder architecture that mainly utilizes the self-attention mechanism to extract intrinsic features, has achieved state-of-the-art results in natural language processing (NLP) (Vaswani et al., 2017) and shown great potential to tackle time series problems (Han et al., 2023). Later, transformer was applied to computer vision (Dosovitskiy et al., 2021), and has also demonstrated significant advantages. For instance, transformer can model long-range interactions in a computationally efficient way, learn more general and robust representations than the local patterns from convolutions, provide a more general architecture for various fields and performs well with large-scale training datasets (Xu et al., 2022).

To benefit from these mentioned advantages, there are a few preliminary works integrating transformer into spatial downscaling. Guo et al. (2023) performed the downscaling process to improve the sub-seasonal prediction skills of summer precipitation in Chongqing through an ensemble model integrated from three deep learning algorithms, including Transformer, sequence to sequence (Seq2Seq) and deep neural networks (DNN). Gerges et al. (2022) developed a novel transformer-based spatial downscaling method Bayesian for daily average wind speed, which consists of attention-based input grouping (AIG) for implicit feature reduction, transformer, and uncertainty quantification implemented based on the Monte-Carlo dropout sampling. Later, they presented a novel deep learning framework WCT (wind convolutional neural networks with transformers) for spatial downscaling of daily average wind speed at a local station level using GCM simulations, which consists of multi-head CNNs, followed by stacked transformers, and an uncertainty quantification component based on Bayesian inference (Gerges et al., 2023). The downscaling performance of the proposed WCT framework has been validated with experimental datasets, and outperformed their previous work AIG-Transformer.

These works are the first steps to introduce transformer into spatial downscaling, and still confronted with many challenges. For instance, the paired datasets with low and high spatial resolution in real scenarios are scarce, and the utilization of more training data by transformers poses a greater challenge to the construction of domain-specific large-scale datasets, and pre-training models (Aleissaee et al., 2023). Besides, recent advantages of combining CNN and transformer (Han et al., 2023), hierarchical transformer (Cai et al., 2023), etc. for super-resolution and other vision tasks can bring ideas for performance improvement of downscaling by feature complementary across scales. Furthermore, the transformer-based models are computationally extensive due to the self-attention mechanism, highly effective and efficient transformer models with low resource cost for spatial downscaling are desired. The model compression and acceleration techniques, e.g., pruning, quantization, knowledge distillation, can still be applied, while additional insights on how to identify redundancy in transformer models are needed (Han et al., 2023).

6.3. Data fusion for downscaling

Currently, a variety of downscaling schemes have been proposed in the context of Earth observations, which however mainly focus on remote sensing images and are partly applicable to gridded meteorological data that reflect quantitative features of variables. These methods generally fall into three categories, i.e. super-resolution, spatio-spectral fusion (SSF), and spatio-temporal fusion (STF), according to the utilization of spectral and temporal information (Sdraka et al., 2022). The SR-based methods are the mainstream for meteorological data downscaling using DL at present, which have been exhaustively discussed in this paper. SSF involves the combination of images with different spatial and spectral resolutions to generate an image with enhanced spatial resolution. It hardly works directly for meteorological data downscaling with derived data products. However, the predictors from different spectral bands, e.g. the optical/thermal and microwave fusion, and the active and passive microwave fusion might be performed to generate specific downscaled results (Sun et al., 2019). STF exploits additional temporal differences, and aims to integrate images of high spatial but low temporal resolution (HSLT) with images of low spatial but high temporal resolution (LSHT). It may help to generate downscaled results with high spatial resolutions by fusing data products derived from satellite constellations for meteorological observations.

Satellite constellation and multi-sensor data fusion, especially the STF is a very promising approach to improve data gaps and resolutions (Sagan et al., 2020). The various instruments equipped onboard different satellites obtain observations with different temporal and spatial resolutions. These satellite remote sensing observations can be used subsequently to inverse meteorological variables. For instance, sea surface wind (SSW) can be obtained from multi-source active and passive sensors with different transit time, swath width and spatio-temporal resolutions, including microwave scatterometers (e.g., HY-2A/HSCAT, Oceansat −2/OSCAT, Metop-A/ASCAT, and Metop-B/ASCAT), microwave radiometers (e.g. F15/SSMI, F17/SSMI, GCOM-W1/AMSR2, and Coriolis/Windsat), etc. Higher spatial resolution observations required are often limited by temporal revisit times to capture quickly changing dynamics (Sagan et al., 2020). On the

contrary, observations with higher temporal resolution typically exhibit coarser spatial resolution. STF can be applied to take advantage of synergistic observations (Zou et al., 2017).

Specifically, with regard to the deep learning-related development, the focus of this review, there are only a few studies in downscaling based on data fusion. Gavahi et al. (2023) developed a general framework that could simultaneously merge and downscale multiple precipitation products to improve the accuracy of quantitative precipitation estimation. Thiria et al. (2023) attempted to downscale sea surface currents through the fusion of heterogeneous observations, i.e. low-resolution sea surface high data and mesoscale resolution sea surface temperature data using an ensemble of CNNs. More state-of-the-art DL-based downscaling models specifically for STF of RS images reviewed by Sdraka et al. (2022) can be referred to, however, they need adaptations (e.g., transform of inputs and outputs, missing value and boundary handling, loss functions, evaluations) towards gridded meteorological data.

In addition, a potential direction to enhance spatial resolution is to combine super-resolution with data fusion. Kim et al. (2023) tended to simultaneously perform error correction and improvement of spatial resolution, and proposed a deep learning framework for multi-source data fusion and super-resolution for generative adversarial network-based spatiotemporal dependency learning to produce accurately downscaled sea surface temperature. The experimental results have shown improved accuracy performance with this proposed combination framework.

6.4. Physical-informed downscaling

DL-based methods have the unique capability to implicitly capture the intricate physical relationships and distributions within meteorological variables. Additionally, the relationship and the distribution can be learned from meteorology and physics, coming out with physical constrained differential equations, such as the motion equation, continuity equation, energy equation, and state equation, etc. Yu et al. (2022b). These physical laws and principles can then be harnessed to fine-tune downscaling results or seamlessly integrate them into the models. Consequently, a significant research focus lies in incorporating these physical constraints as prior knowledge into data-driven DL models, thereby enhancing performance and interpretability. These methods are known as physical-informed deep learning (PIDL). According to the way that physical information is integrated, they can be further classified into the following six distinct integration methods.

- Physics-informed input: Incorporating physics-related information as inputs to the model, such as known physical parameters, physical boundary conditions, or other relevant physics information. By effectively utilizing this physical information, the model can improve its predictive performance (Yasuda et al., 2021).
- Physics constraints: Adding physics constraints to the loss function of a deep learning model ensures that the predicted results align with the laws of physics. This can be achieved through regularization terms, constraint conditions, or custom loss functions. For example, in computational fluid dynamics, a loss term based on the Navier–Stokes equations can be incorporated to ensure that the predicted velocity field satisfies the fundamentals of fluid mechanics (Wessels et al., 2020).
- Physics-based regularization: Using physics knowledge to design regularization terms that constrain the learning process of the model. For instance, regularization terms can be designed based on the boundary conditions of partial differential equations to ensure that the model behaves in accordance with physical constraints at the boundaries (Lazovskaya et al., 2021).
- Multi-physics integration: Integrating information from multiple physics domains into a single model to solve problems involving the coupling of multiple physical phenomena. For example, integrating heat conduction and fluid dynamics models into one model to simulate the behavior of a liquid during a heating process (Kubeyev, 2023).
- Data assimilation: Combining physics models with actual observational data to enhance the model's predictive performance. This can be achieved by comparing the observed data with the model's output and using data assimilation techniques to correct the model's predictions (Feng et al., 2023b).
- Physics-informed network architectures: Designing network architectures that incorporate physics knowledge to guide the learning process of the model. For instance, in the case of gridded meteorological data, specific convolutional kernels can be designed within CNNs to capture specific physical features more effectively (Feng et al., 2023a).

As for the meteorological data downscaling, integrating physics into DL networks remains challenging, and there are only a few preliminary papers in this field. Yasuda et al. (2021) proposed a physics-informed method for the near-surface temperature downscaling. Their method is implemented by simply adding auxiliary variables as input, and such methods have been discussed in Section 4.3. Chen et al. (2022) further proposed a physical-informed network for HR urban surface temperature estimation. The network contains two input branches, the global physics feature perception branch and the local urban surface perception branch for multiple atmospheric forcing variables and local geometric information, constraining the estimations by the relevant physical laws implied by the physical information of the two branches. However, these methods do not explicitly incorporate physical constraints.

So far, all these meteorological data downscaling methods based on PIDL have adopted the approach of incorporating additional physical-informed input into the networks. These existing methods integrate the related variables introduced in Section 4.3 into the networks as soft constraints. In addition, the coupling downscaling methods for other satellite datasets may also provide new insights. Sun et al. (2022) explored different serial and parallel strategies to couple the learning-base methods and the physical or semiphysical (PH) models. In the parallel mode, the downscaled HR results can be integrated from both methods using arithmetic average, geometric average, weighted average etc. In the serial mode, the initial estimates from the PH models can serve as one of the independent input variables in the learning algorithm, in reverse order, learning methods can be used to perform preliminary steps such as data interpolation for the physical or PH models.

From a broader perspective, further research is expected to apply more effective ways of integrating physics into DL networks. One potential direction is to incorporate explicit physical constraints into the loss function (Jiang et al., 2020) or regularization terms (Lazovskaya et al., 2021) of the network. This ensures that the predicted results align with the laws of physics. Additionally, leveraging physics-based regularization techniques, such as designing regularization terms based on boundary conditions or physical principles, can further enhance the model's adherence to physical constraints.

There are also pioneering studies on PIDL, whose special networks or loss functions are constructed based on characteristic equations, boundary conditions, and other inherent relationships and constraints in different domains, e.g., fluids, quantum mechanics, reaction–diffusion systems demonstrated in Raissi et al. (2019). These PIDL techniques have been applied to image SR in various scenarios. For example, Jiang et al. (2020) proposed a novel physics-constrained DL model for turbulent flows SR, which consistently improves the recovery of key physical quantities of interest compared to competing baselines. Wang et al. (2020a) developed a physics-informed neural network SR method for advection–diffusion models, where an additional loss

term with physical constraints is constructed. These approaches provide potential solutions for physics-informed downscaling.

Furthermore, exploring multi-physics integration (Kubeyev, 2023) approaches can be beneficial for capturing the complex interactions between different physical phenomena in meteorological data. By integrating information from multiple physics domains, such as atmospheric dynamics, thermodynamics, and radiation, into a single model, it becomes possible to simulate and downscale meteorological variables more accurately.

6.5. Downscaling model generalization

The generalization capacity of downscaling models is a significant challenge and a major concern, and should be carefully assessed. This refers to applying the trained downscaling models to the unseen testing data with the predictors which formally depict the same meteorological variable as that used during training, but deviate from the training data in terms of certain properties. For instance, the temporal context varies in the widely used dataset construct pattern, as stated in Section 4.4 (Höhlein et al., 2020).

For the DL-based downscaling models, as the depth of network increases, its nonlinear fitting ability is stronger. This provides higher accuracy while bringing the overfitting issue that leads to the inadaptability of the model from unseen data. Although there are tricks to alleviate overfitting during training, such as regularization (Zhang et al., 2021a), dropout (Liu et al., 2019a), batch normalization (Garbin et al., 2020) etc., it can hardly meet the requirements of downscaling in the conditions where the training data is rare. Besides, the changing environment and data observation scenes in diverse conditions (e.g., geo-locations, time, sensing platforms) may cause data distribution shifts (Tong et al., 2020; Lunga et al., 2021). Therefore, the DL-based models trained on a set of samples are generally restricted to a specific domain, for instance, an area or dataset (Zhang et al., 2021d; Sha et al., 2020b), or under certain meteorological conditions (Khorrami et al., 2021). They should not be expected to consistently yield satisfactory performance when directly applied to unseen contexts (Xiang et al., 2023).

To enhance generalization performance, an obvious approach is to collect massive amounts of data that cover almost all possible meteorological contexts, which is costly and difficult to achieve since some observations are sparse or unavailable (Khorrami et al., 2021). However, compared with some other fields, there is a large amount of data generated from numerical models which can be exploited for the downscaling model training (Rampal et al., 2022). In addition to augmenting data, transfer learning is a general practice to enhance model generalization and improve performance. It refers to using the knowledge acquired from a task with a larger number of samples for performing a task in a similar but different domain that may have few or even no data (Yosinski et al., 2014). Transfer learning has been considered a plausible direction to solve the issue of downscaling model generalization. For instance, Sha et al. (2020a) performed semi-supervised transfer learning for UNet, and the resulting UNet-AE could adapt to new spatial domains in downscaling problems. Wang et al. (2021) performed transfer learning to explore the performance of the propose SRDRN model for downscaling over data sparse regions. Available evidence suggests that the temporal generalization is more easily obtained, i.e. the models learned for a certain period have been generally applied successfully to new periods (Dujardin and Lehning, 2022). Moreover, the related efforts to enhance model generalization are divided into the following aspects, including weakly supervised learning, optimal model and feature selection, semi-supervised domain adaptation, deep metric-based methods for few-shot learning, and meta-learning with deep metric embedding (Lunga et al., 2021). It is important to note whether the improved downscaling models are capable of capturing non-stationarity (Wu et al., 2022), a longstanding issue in statistical downscaling (Vandal et al., 2017).

6.6. Downscaling uncertainty quantification

Downscaling uncertainty refers to the uncertainty in the downscaled results, mainly capturing aleatoric uncertainty and epistemic uncertainty. The former accounts for the randomness and noise in the input observed data, while the latter depicts the uncertainty from downscaling models (Vandal et al., 2018). It may lead to an ensemble of possible but varied downscaled outputs (Leinonen et al., 2021), which can be characterized with statistical methods, e.g. the commonly used means, variances, confidence intervals (Khan et al., 2006).

Although there exists inherent uncertainty in image SR reconstruction, most methods produce a single deterministic output from the conditional distribution of HR data for a given LR input. Similarly, the existing deep learning-based downscaling methods for meteorological data are generally deterministic models, and hardly provide uncertainty (Wang et al., 2021; Sha et al., 2020a). It is generally acceptable to generate a plausible solution with enhanced resolution for natural images (Leinonen et al., 2021). However, in the field of meteorological data downscaling, it is important to understand and quantify uncertainty, which helps to characterize the conditional distribution of outputs and give hints for reasonable data use in special scenarios, e.g. extreme events (Stengel et al., 2020). Therefore, for a given low-resolution input, an ensemble of predictions with probabilistic estimations, in addition to a single deterministic output from the conditional distribution of high-resolution data is desired (Leinonen et al., 2021).

Researches have indicated that the uncertainty increases, especially in the condition of extreme events or increasing downscaling factor (Mu et al., 2020b), and the uncertainty of downscaling mainly stems from the ill-posed problem of downscaling, the spatial and temporal non-stationarity of the changing climate system, as well as sparse occurrences and skewed distributions (Vandal et al., 2017). However, the uncertainty issue has been largely ignored in the earlier endeavors of DL-based downscaling to improve the spatial resolutions.

Uncertainty quantification (UQ) in neural network predictions has been paid attention in science and engineering, including medical diagnosis and robotics, among others. The uncertainties of deep neural networks mainly include model uncertainty and data uncertainty, and Bayesian approximation and ensemble learning are the most widely-used UQ methods currently (Abdar et al., 2021). The Bayesian DL models quantify uncertainty by using the distribution over model weights or learning a direct mapping to probabilistic outputs directly. In the ensemble learning methods, the predictions of several different deterministic network models are combined to improve generalization capability and reduce uncertainty. The ensembles that belong to the mixture models can be employed to quantify the uncertainty. In addition, the uncertainty quantification can also be performed on a single deterministic network, but generally with test-time data augmentation (Gawlikowski et al., 2022). For example, Sha et al. (2020a) observed the performance of predictions, given outliers and ambiguous inputs, while Paschali et al. (2018) evaluated the uncertainty with crafted adversarial samples.

For the downscaling task, a few studies attempt to quantify uncertainty mainly based on the recent developments in Bayesian DL. Vandal et al. (2018) applied Bayesian DL with normal and lognormal likelihoods for UQ on DeepSD, a DL-based statistical downscaling model applied to precipitation downscaling. Leinonen et al. (2021) presented a conditional SR GAN to create an ensemble of predictions for a single input, and the recurrent layer in the generator can be understood as performing a Bayesian update on the ensemble member. Mu et al. (2020a) presented a multi-scale Bayesian convolution network that replaces the conventional convolutional layer with the Bayesian convolution layer, thus addressing the uncertainty issue.

UQ facilitates better understanding and decision making of downscaling results. Further attempts to provide solutions for reducing uncertainty are also crucial. Sha et al. (2020a) summarized the related

directions, including state-of-the-art adversarial training that helps deal with the input space uncertainty (Zhu et al., 2017), the predecessor denoising step to convert the input to a clean LR which could be integrated into the downscaling pipeline as an end-to-end network, and the loss considering the physics-based constraints (Qian et al., 2019). Besides, ensemble inference from a set of HR reconstruction candidates which might come from multiple downscaling paths is also an important and effective way to reduce the downscaling model prediction uncertainty (Mu et al., 2020b).

7. Conclusions

To fulfill the refined data needs of regional and local studies, in addition to developing HR sensors (e.g. synthetic aperture radar systems) and numerical models, downscaling has long been an important process to derive finer-resolution estimates from LR data. The widely used traditional downscaling methods mainly include dynamic downscaling and statistical downscaling. The former category of methods is with explicit physical principles, less influenced by observation data, but computationally expensive. The latter is data-driven and can reduce the computation significantly. In recent years, due to the ability to learn complex nonlinear feature representations, DL has been introduced into the statistical downscaling, but is still in quite an early stage. Therefore, this work attempts to give a systematic review of recent developments in meteorological data downscaling by incorporating cutting-edge DL techniques.

In this review, we first introduce the downscaling background and give an overall evolution summary, including the concepts, and methods with characteristics in categories. Subsequently, we explain the role of DL in statistical downscaling, and analyze image SR in computer vision, a task similar to statistical downscaling which significantly inspired the development of DL-based statistical downscaling. Currently, typical DL network architectures, including CNN to capture the spatial dependencies of meteorological variables, RNN to reveal the temporal states from time series, and GAN to facilitate the reconstruction of high-frequency details have been involved in downscaling. Residual modules and attention mechanisms have also been integrated as critical structures to enhance performance.

In addition to the fundamental deep architectures, we thoroughly analyze the major specific issues towards the meteorological data downscaling, including downscaling scale factor, spatial–temporal correlation, auxiliary variables for downscaling, and accessibility of paired datasets for training. Furthermore, in order to make up for the lack of a relatively unified data set and facilitate the development, validation and comparison of subsequent relevant research for the community, we give a summary of the downscaled variables, the corresponding LR and HR datasets, the available downscaling toolsets, and evaluation metrics of DL-based meteorological data downscaling methods.

Current studies have demonstrated the feasibility and superior performance of DL in the field of meteorological data spatial downscaling. However, great challenges still need to be addressed. Firstly, acquiring HR data can be challenging, making the unsupervised learning based downscaling to derive HR data without paired training datasets highly practical. Secondly, it is crucial to carefully assess and enhance the generalization capacity of downscaling models across diverse contexts. Besides, it is important to quantitatively evaluate and reduce uncertainty in DL-based downscaling methods. Potential development of methods are also desired, e.g. transformer for downscaling, data fusion for downscaling, and incorporating physical information into data-driven DL-based downscaling models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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